



UNIVERSITY OF AMSTERDAM
Economics & Business

MSc Econometrics | Track: Data Analytics

xG Beyond Event Data: Spatiotemporal Tracking and Machine Learning in Football Shot Quality Modelling

Rafeh Islam (12700479)

Supervisor: Hans van Ophem

Second reader: Simon Trimborn

Abstract

Expected Goals (xG) has become central to football analytics, yet most publicly available models rely on event data alone, ignoring player positions at the moment of the shot. This thesis investigates to what extent spatiotemporal tracking data improves xG prediction and which machine learning approach benefits most. Using 7,885 non-penalty shots from five competitions with StatsBomb 360 freeze-frame coverage, five model families are evaluated across four feature conditions and benchmarked against StatsBomb's commercial model via DeLong tests with Holm-Bonferroni correction. Freeze-frame tracking improves AUC from 0.826 to 0.841 over the event-data baseline. The best model, a tuned ElasticNet logistic regression with combined tracking and temporal features, achieves 98.0% of StatsBomb's discriminative performance. SHAP analysis reveals that the marginal value of each data stream depends on model architecture. This study contributes novel temporal possession-sequence features and a statistically robust and reproducible benchmarking framework using open data.

Keywords: Expected Goals, Spatiotemporal Tracking, Machine Learning, StatsBomb 360, Data Analytics, Football Analytics

Statement of Originality

This document is written by Rafeh Islam, who declares to take full responsibility for the contents of this document. I declare that the text and the work presented in this document are a reflection of my own efforts and conform to all rules regarding academic integrity. All sources and tools used in its creation, including any AI-based applications, are appropriately acknowledged and referenced. UvA Economics and Business is responsible solely for the supervision of completion of the work and submission, not for the contents.

Generative AI tools (Anthropic, 2025) were used to assist with editing, proofreading, and structuring during the writing process. All analytical work, research design, modelling, statistical testing, and interpretation of results are entirely the author's own.

Contents

1	Introduction	4
2	Literature Review	7
2.1	Statistical Foundations of Football Modelling	7
2.2	Machine Learning Approaches to xG Modelling	8
2.3	Incorporating Tracking and Spatiotemporal Data	9
2.4	Extensions, Player Quality, and Match-Level Applications	10
2.5	Positioning This Study	11
3	Data	13
3.1	Data Source	13
3.2	Sample Construction	14
3.3	Competition-Level Descriptives	15
3.4	Shot Outcome Distribution	15
3.5	Key Descriptive Statistics	16
3.6	Body Part and Play Pattern Distribution	17
3.7	Descriptive Comparison: Goals vs Non-Goals	18
3.8	Feature Correlation Structure	20
3.9	Shot Location Distribution	21
3.10	Temporal Feature Overview	22
4	Methodology	23
4.1	Research Design	23
4.2	Feature Engineering	24
4.2.1	Baseline Spatial Features	24
4.2.2	Shot Context Features	24
4.2.3	Tracking-Derived Goalkeeper Features	25

4.2.4	Tracking-Derived Defender Features	25
4.2.5	Temporal Possession Features	26
4.2.6	Interaction Terms	26
4.3	Model Families	27
4.4	Hyperparameter Tuning	28
4.5	Evaluation Metrics	30
5	Results	32
5.1	Core Finding: The Marginal Contribution of Spatiotemporal Data	32
5.2	Full Model Leaderboard	33
5.3	SHAP Feature Importance Analysis	36
5.3.1	Cross-Scenario SHAP Analysis	36
5.3.2	ElasticNet Logistic Regression Coefficient Analysis	40
5.4	Model Diagnostics	42
5.4.1	ROC Analysis	42
5.4.2	Calibration Analysis	43
5.4.3	Precision-Recall Analysis	44
5.5	Performance Breakdown	44
5.5.1	Header Gap Analysis	44
5.5.2	Competition and Gender Subgroup Analysis	46
5.6	Bundesliga Betting Case Study: Poisson Aggregation Framework	47
6	Conclusion	52
6.1	Summary of Findings and Concluding Remarks	52
6.2	Limitations and Directions for Future Research	53

Chapter 1

Introduction

As the 2026 FIFA World Cup unfolded across the United States, Canada, and Mexico, football's commercial scale has reached unprecedented heights. FIFA projects revenues of approximately \$13 billion from the four-year commercial cycle culminating in the tournament, up from \$7.5 billion in the cycle ending with the 2022 Qatar edition (FIFA, 2024).

This financial scale underscores the commercial stakes of modern football, with data being central to how the game's major stakeholders operate. Clubs and federations increasingly rely on quantitative models of player and team performance for tactical analysis, scouting, and transfer valuation. Broadcasters, meanwhile, increasingly embed live probabilistic metrics into match coverage, while betting companies depend on the same underlying data to price risk accurately. Football accounts for the largest share of any sport within the global betting industry, valued at over \$110 billion in 2024 and projected to reach \$236 billion by 2033 (Grand View Research, 2025). Odds compilers, quantitative traders, and betting syndicates worldwide depend on accurate probabilistic models of match and shot outcomes to gain any edge over the market.

At the centre of this analytical ecosystem sits Expected Goals (xG), a metric that has moved from niche analytics circles to mainstream broadcast graphics in less than a decade. Where once commentators described chances in qualitative terms, audiences watching the 2026 World Cup saw xG figures displayed in real time alongside the score. At its core, xG is a probabilistic estimate of whether a given shot would result in a goal, based on the characteristics of the shot: its location, the body part used, and the circumstances under which it was taken. Aggregated across a match, xG provides a measure of the quality of chances created and conceded, enabling

more meaningful performance comparisons than raw scorelines allow (Brecht and Flepp, 2020).

Yet, despite xG’s ubiquity, a fundamental limitation persists in many publicly available models: they are built from event data alone, capturing only what happens to the ball, without knowing where the players are. When a striker shoots from eighteen metres, a standard xG model incorporates the distance and angle, but not whether defenders are blocking the path to goal, where the goalkeeper is positioned relative to the goal, or how many attackers are in support. This spatial blindness represents a systematic omission of the information that determines whether a shot is genuinely threatening.

The proliferation of tracking technology in professional football is beginning to close this gap. Modern stadiums are equipped with multi-camera optical tracking systems that record player positions up to 25 times per second. GPS units and inertial measurement units worn by players capture accelerations, distances, and movement patterns at the individual level. These technologies collectively generate hundreds of millions of data points per match, enabling a level of spatial and temporal resolution that was unimaginable even a decade ago.

StatsBomb has made a subset of this spatial information publicly available through their 360 data product, which captures the positions of all visible players at the exact moment of every recorded event, including every shot. This freeze-frame data makes it possible to build xG models that incorporate not just where the shot was taken from, but what the defensive and goalkeeper configuration looked like at the moment it was taken. The extent to which this additional spatial context translates into better predictions, and which machine learning methods are best placed to exploit it, is the central question this research addresses.

Specifically, this thesis investigates the following research question:

To what extent does incorporating spatiotemporal tracking data improve xG (Expected Goals) prediction, and which machine learning approach benefits most?

To answer this, 7,885 non-penalty shots from five competitions with complete 360 freeze-frame coverage are used to systematically compare five machine learning model families (logistic regression, random forest, XGBoost, LightGBM, and CatBoost) across four feature conditions: a pure event-data baseline, a tracking-enhanced specification, a temporal possession-sequence specification, and a combined feature set of 82 variables. All models are hyperparameter-tuned and benchmarked against StatsBomb’s own commercial xG model on the same test set, with statistical significance of performance differences assessed via paired DeLong tests with Holm-

Bonferroni correction.

The main findings are as follows. Adding freeze-frame tracking features to a logistic regression baseline improves AUC from 0.826 to 0.841, a lift of +0.015 achieved with no change in model complexity. The best model, a tuned ElasticNet logistic regression with combined tracking and temporal features, achieves an AUC of 0.850, representing 98.0% of StatsBomb’s discriminative performance, with the residual gap plausibly reflecting StatsBomb’s substantially larger training set and more granular model architecture (StatsBomb, 2022). All five model families show statistically significant AUC improvements from baseline to tuned specification. SHAP analysis reveals that temporal features dominate in the best logistic regression, while tracking features dominate in Random Forest despite identical feature sets, indicating that the marginal value of each data stream depends materially on model architecture. A Poisson aggregation case study on the 2023/24 Bundesliga season demonstrates a complete pipeline from shot-level xG to match outcome probabilities.

The remainder of this thesis is structured as follows. Chapter 2 reviews the relevant literature on xG modelling, machine learning approaches in football analytics, and the emerging use of tracking data. Chapter 3 describes the StatsBomb dataset and descriptive statistics. Chapter 4 details the feature engineering process and modelling methodology. Chapter 5 contains the empirical results and Chapter 6 concludes with a discussion of contributions, limitations, and directions for future research.

Chapter 2

Literature Review

2.1 Statistical Foundations of Football Modelling

Probabilistic modelling of football outcomes predates the machine learning era by decades. The foundational statistical framework for football analytics was established by Dixon and Coles (1997), which models the two teams' scores as independent Poisson processes with a correction for dependence in low-scoring outcomes, using English league data from 1992–1995. This work demonstrates that team-specific attack and defence parameters can generate match outcome probabilities with positive betting market returns. While operating at the match level rather than the shot level, it establishes the core principle that football outcomes are probabilistic, and that structured statistical models can extract genuine signals from noisy match data.

Shooting in football has long been recognised as the ultimate determinant of match outcomes, motivating decades of research into the factors that make a shot more or less likely to succeed (Pollard and Reep, 1997). xG emerged as a methodological response to the limitations of traditional goal-based performance metrics, where a shot is a far more frequent event than a goal, offering a more stable basis for accounting for the substantial randomness inherent in football outcomes (Noordman, 2019; Brechot and Flepp, 2020). This distinction between attempts and actual goals underlies the widespread adoption of xG as an alternative measure of team performance to final score.

xG provides a probabilistic estimate of the likelihood that a given shot results in a goal, based on characteristics related to the shot. From a modelling perspective, xG is typically framed as a binary classification problem, which has motivated the application of a wide range of statistical and machine learning techniques (Mead et al., 2023; Hewitt and Karakuş, 2023; Corsaro et al.,

2025). Pollard et al. (2004) is among the earliest to estimate the probability of a shot resulting in a goal using logistic regression, incorporating distance from goal, shooting angle, and the space available from the nearest opponent at the moment of the shot. The study notes that additional factors likely influence shot quality beyond these three variables.

Rathke (2017) provides an early systematic attempt to quantify shot conversion probability, analysing 18,218 shots across the English Premier League and German Bundesliga using Opta event data, partitioning the pitch into eight discrete spatial zones and assigning empirical conversion rates to each. This zonal framework confirms that a combination of distance and angle dominates shot conversion probability, with central close-range zones yielding considerably higher conversion rates than shots taken from a distance. The study explicitly notes that shot-type variation, including passes, through balls, and set-play context, was not examined in detail, a gap that motivates the features developed in the present study.

2.2 Machine Learning Approaches to xG Modelling

Football shot quality plausibly involves nonlinear relationships and complex interactions, for instance between defensive pressure and shot angle, or between distance and goalkeeper positioning, motivating exploration of more flexible machine learning architectures capable of learning such interactions directly from the data (Chen and Guestrin, 2016).

Beyond logistic regression, a range of machine learning methods have been applied to xG estimation. Mead et al. (2023) provides a comprehensive ML benchmarking study comparing logistic regression, random forests, MLP (Multi-Layer Perceptron), AdaBoost, and XGBoost across the top five European leagues using Wyscout data from the 2017/18 season. This study finds that incorporating non-spatial macro-level features, including player market value and goal differential, meaningfully improves predictive performance for future match outcomes, ranking amongst the most influential features in these models. However, goal differential may capture team quality rather than psychological state, as stronger teams are more likely to be ahead in matches. The present study takes a different approach. Rather than maximising predictive accuracy using any available signal, the objective is to isolate the marginal contribution of shot-level spatial and temporal context specifically. Match-state variables such as goal differential are therefore deliberately excluded, not because they lack predictive value, but because they fall outside the shot-quality framework investigated here.

Corsaro et al. (2025) provides an independent validation of ML family rankings using 100,205

Serie A shots spanning ten seasons from Understat, finding that a LightGBM specification best approximates its commercial xG model, while its feedforward neural network subtly outperforms it. This contrasts with Méndez et al. (2023), which applies a feedforward MLP to 16,042 StatsBomb shots and finds the model underperforms StatsBomb’s benchmark on accuracy and recall, but outperforms it on precision and F1-score. Together these studies suggest that neural network performance relative to commercial xG benchmarks is sensitive to the evaluation metric and dataset. Cavus and Biecek (2022) similarly compares XGBoost, Random Forest, LightGBM, and CatBoost across 315,430 shots from the top five European leagues, finding Random Forest to be the best-performing model.

Hewitt and Karakuş (2023) makes an important methodological contribution demonstrating the power of feature engineering within an event-data framework. Using 15,575 shots, this study shows that expanding its baseline logistic regression specification substantially improves its correlation with StatsBomb’s xG values, from $r = 0.659$ in an initial baseline to a final tree-based model reaching $r = 0.902$. This demonstrates that feature engineering can close the performance gap with commercial models.

2.3 Incorporating Tracking and Spatiotemporal Data

While traditional xG models rely on event data (Rathke, 2017; Noordman, 2019; Mead et al., 2023), recent advances in data collection have enabled the use of spatiotemporal tracking data, which records the continuous positions of all players and the ball throughout a match at frequencies of up to 25 times per second (25 Hz). This high-frequency data captures off-ball movement, defensive structure, and dynamic positioning at every instant, offering a more detailed representation of the game state. Commercial providers such as Opta, Wyscout, and StatsBomb collect and incorporate it into their own xG models and license it directly to clubs, betting firms, and broadcasters.

A foundational contribution in this area is made by Lucey et al. (2015). Using nearly 10,000 shots from a full professional league season, this study demonstrates that strategic spatiotemporal features derived from continuous player tracking significantly improve shot prediction. The fundamental limitation of event-based xG models lies in their static representation of inherently dynamic spatial interactions, particularly their inability to capture defensive pressure and player positioning at the moment of the shot (Lucey et al., 2015). The key innovation of this study is the defensive triangle descriptor, which measures whether defenders occupied the direct shooting lane from shooter to goalposts. Eggels (2016) develops a similar tracking-based

approach using 10 Hz positional data, extracting the number of attackers and defenders positioned between the shooter and goal, and the Euclidean distance of the goalkeeper and nearest defender to the attacker.

Spearman (2018) provides broader conceptual grounding for the temporal dimension of shot quality, arguing that the value of a scoring opportunity depends not only on the instantaneous spatial configuration at the moment of the shot but on the dynamics of play leading up to it. Similarly, Lucey et al. (2015) also examines events within a fixed window prior to the shot, offering an early precedent for the 10-second pre-shot temporal window. However, this evaluation reports only average prediction error without decomposing feature-level importance or coefficients, limiting insight into which specific spatiotemporal features drive the reported improvement.

A relatively comparable study to the present thesis is Anzer and Bauer (2021), which develops a goal scoring probability model using synchronised, proprietary tracking data across 105,627 Bundesliga shots over seven seasons. The best-performing XGBoost model achieves a higher AUC than an event-only baseline, and SHAP analysis identifies goalkeeper distance to goal and maximum individual pressure difference as the most important tracking features alongside raw distance and angle. Anzer and Bauer (2021) notes ball height tracking as a key source of inaccuracy in its header sub-model, even with z-coordinate ball position included, since small variations in ball height at contact substantially affect shot placement but are difficult to track reliably from video. This limitation is directly relevant here, since StatsBomb open data provides only planar (x, y) positions with no vertical information.

2.4 Extensions, Player Quality, and Match-Level Applications

Beyond shot-level modelling, several studies have explored extensions to the xG framework. Noordman (2019) develops three progressively complex frameworks for converting shot-level xG into match outcome probabilities, using 146,540 shots from Wyscout enriched with FIFA 19 player skill indices. Adding player quality marginally improves shot-level AUC. This study explicitly identifies the incorporation of spatiotemporal tracking data, rather than event data alone, as a natural direction for future research, a gap the present study directly addresses. Eggels (2016) similarly aggregates shot-level expected goals to the match level to generate expected outcomes, using tracking data at 10 Hz frequency.

Beyond prediction accuracy, both Noordman (2019) and Eggels (2016) situate their match-level

frameworks within a betting context. Noordman (2019) finds that expected-points rankings outperform actual league standings as a predictor of final position and speculates that this approach could similarly outperform bookmaker expectations over time. Baboota and Kaur (2019) applies gradient boosting to English Premier League match outcome prediction across 11 seasons, finding that engineered features, particularly a dynamic team form metric, dominate raw statistics, though its best model falls narrowly short of bookmaker accuracy on the ranked probability score. The narrow gap between machine learning models and market-implied probabilities across these studies motivates the betting market comparison developed in Section 5.6 of the present thesis.

A related strand of literature explores player-adjusted extensions of xG, aiming to isolate individual finishing skill from underlying shot quality. Hewitt and Karakuş (2023) illustrates the sensitivity of this approach by showing that retraining an xG model on a single elite player (Lionel Messi) inflates accumulated xG across the remaining sample by 347 goals, highlighting a strong reference-player dependency that restricts cross-competition generalisability. Relatedly, Davis and Robberechts (2024) shows that standard xG models exhibit a persistent bias underestimating elite finishers’ true skill. Bayesian hierarchical frameworks offer an alternative approach by isolating player and positional effects on xG. Scholtes and Karakuş (2024) finds that positional effects attenuate significantly when rich spatial predictors are incorporated, supporting an emphasis on rich feature engineering. Motivated by these insights, the present study intentionally omits player identity features. While including player fixed effects or finishing traits could capture additional predictive signal, they introduce reference-player dependencies and limit cross-dataset generalisability. Consequently, the models estimated here isolate the intrinsic quality of a shot given its spatial and temporal context, rather than maximizing individual goal-prediction accuracy. This also keeps the benchmark comparison fair, since StatsBomb’s own published methodology confirms its models rely exclusively on shot-level features (StatsBomb, 2022).

2.5 Positioning This Study

The literature reviewed above traces a clear progression from simple geometric shot models to machine learning approaches, tracking-enhanced xG, and extensions toward match outcome prediction, betting, and player evaluation. Despite this progress, several important gaps remain.

First, while a number of studies show that tracking data improves xG prediction, these improvements are typically reported for a single model, and it remains unclear whether tracking data

benefits all model classes equally or whether certain algorithms are better equipped to exploit its additional information. This thesis addresses this by evaluating five model families under identical feature conditions ranging in complexity, providing evidence on whether the marginal value of tracking and temporal data is model-dependent.

Second, existing work incorporating spatiotemporal context has largely focused on the instantaneous moment of the shot, with comparatively little attention paid to the temporal dynamics of the possession that created it. This thesis extends the tracking-based xG literature in this dimension, constructing dedicated possession-chain and pre-shot temporal features and evaluating them alongside freeze-frame spatial data as a distinct predictor category.

Third, many studies that incorporate tracking data use proprietary datasets (Anzer and Bauer, 2021). Where open data is used, direct benchmarking against a commercial provider suffers from one of several limitations: it is restricted to a single model specification, it evaluates similarity to the provider’s own xG values rather than independent performance against realised outcomes (Scholtes and Karakuş, 2024), or it relies on threshold-dependent classification metrics without statistical significance testing (Méndez et al., 2023). This makes results difficult to compare across studies or against commercial performance. This thesis addresses this by systematically benchmarking models independently against StatsBomb’s proprietary xG estimates, providing an independent comparison of open-data models with a commercial benchmark. Corsaro et al. (2025) similarly frames this opacity as a problem for the field, noting that the proliferation of undisclosed provider-specific models “makes this metric unclear and decreases the effectiveness of using xG for investment strategies, decisions, and more” (Corsaro et al., 2025, Sect. 6).

Ultimately, this thesis bridges methodological advances in machine learning with the practical application of tracking and temporal data in football analytics, while contributing to greater transparency in an area of sports analytics still substantially shaped by undisclosed models and data.

Chapter 3

Data

3.1 Data Source

This study utilises the StatsBomb 360 Open Data, a publicly available dataset released via the official StatsBomb GitHub repository. The dataset is particularly well-suited for this study as it provides a combination of event-level data and tracking information, enabling direct comparison between traditional xG models and those incorporating spatiotemporal context. StatsBomb’s datasets have been used in several academic studies (Appiah et al., 2026; Hewitt and Karakuş, 2023; Méndez et al., 2023). The dataset comprises two complementary data streams that are merged for this study. The first is the standard event data stream, which records every on-ball action during a match including attacking and defensive actions, along with additional contextual data such as location coordinates, play pattern, and shot technique. The second is the StatsBomb 360 freeze-frame stream, which captures the spatial coordinates of all players visible in the camera frame at the exact moment of every recorded event. This study uses the freeze-frames attached to goal attempts, combined with events from the preceding 10 seconds. This provides the positions of all players at the instant the attempt is made, enabling the construction of defensive and goalkeeper positioning features that are unavailable in standard event data.

StatsBomb markets its commercial xG model as designed to see “what others miss,” explicitly citing shot impact height and goalkeeper and defender locations captured through this freeze-frame functionality as key differentiators (StatsBomb, 2024), motivating the present study’s use of the same underlying data source to construct comparable spatial tracking features.

3.2 Sample Construction

The dataset covers 323 matches with complete freeze-frame coverage across five competitions: the UEFA European Championship, the UEFA Women’s European Championship, the FIFA Women’s World Cup, the FIFA World Cup, and the German Bundesliga (2023/24 season). These competitions span both men’s and women’s football across international and domestic settings, with variation in playing styles and competitive levels. While pooling these distinct settings introduces potential structural heterogeneity, this broad sample maximizes both sample size and external validity. To address potential subgroup dynamics, Section 5.5.2 provides a disaggregated evaluation grouped by competition and gender.

All shot events are extracted from the event stream and merged with their corresponding freeze-frame data. Following standard practice in the xG literature (Anzer and Bauer, 2021; Hewitt and Karakuş, 2023; Rathke, 2017), penalty kicks are removed from the dataset prior to modelling. Penalties are excluded because they are taken under highly standardised conditions, such as a fixed location of the ball and goalkeeper, and no defenders involved, rendering most spatial features less informative. Additionally, their conversion rate of approximately 64% in this dataset is substantially higher than open-play shots, and including them would artificially inflate model performance metrics. Shots classified under the “Other” play pattern are also excluded, as these correspond predominantly to penalties and a small number of anomalous open-play attempts with unusually high conversion rates. Own goals are excluded as they do not represent intentional shot attempts.

After these exclusions, the final dataset contains 7,885 non-penalty shots across 323 matches. Of these, 757 result in goals, corresponding to an overall conversion rate of 9.6%. Conversion rates in the literature vary by data source and competition, ranging from 10.1% (Appiah et al., 2026) to approximately 11.0% (Eggels et al., 2016) and a somewhat higher 12.5% (Noordman, 2019).

This sample size is comparable in scale to several studies (15,575 shots, Hewitt and Karakuş, 2023; 18,218 shots across the 2012/13 English Premier League and Bundesliga seasons, Rathke, 2017), though considerably smaller than studies using larger commercial datasets (105,627 shots, Anzer and Bauer, 2021; 100,205 shots, Corsaro et al., 2025).

3.3 Competition-Level Descriptives

Table 3.1 presents shot counts, goal counts, and conversion rates broken down by competition. Shot volume varies considerably across competitions, from 881 shots in the Bundesliga to 2,459 in the UEFA European Championship. Conversion rates range from 8.29% (FIFA Women’s World Cup) to 11.35% (Bundesliga), with women’s competitions falling within a similar range to men’s competitions. This consistency across competitions suggests that the mean conversion rate is broadly stable across professional football contexts, though the sample is limited to five competitions.

Table 3.1: Shot and goal counts by competition.

Competition	Shots	Goals	Conv.%
UEFA Euro	2,459	216	8.78%
UEFA Women’s Euro	1,603	165	10.29%
Women’s World Cup	1,556	129	8.29%
FIFA World Cup	1,386	147	10.61%
Bundesliga (2023/24)	881	100	11.35%
Total	7,885	757	9.60%

3.4 Shot Outcome Distribution

Beyond the binary goal/no-goal classification used in modelling, shot outcomes fall into several categories in the StatsBomb dataset. Table 3.2 presents the full distribution. The majority of shots miss the target entirely, with 31.2% going off target and 6.2% classified as wayward. Blocked shots account for 27.3% of all attempts, highlighting the importance of defensive positioning at the moment of the shot. Similarly, saved shots represent 22.6% of attempts, underlining goalkeeper positioning.

Table 3.2: Shot outcome distribution.

Outcome	Count	%
Off Target	2,460	31.2%
Blocked	2,154	27.3%
Saved	1,784	22.6%
Wayward	490	6.2%
Goal	757	9.6%
Post	174	2.2%
Saved Off Target	33	0.4%
Saved to Post	33	0.4%
Total	7,885	100%

3.5 Key Descriptive Statistics

Table 3.3 presents descriptive statistics for the key spatial and tracking variables used in this study. The mean shot distance is 18.1 metres with a median of 17.4 metres, reflecting a right-skewed distribution driven by a tail of long-range attempts. This is closely comparable to (Appiah et al., 2026), which reports a mean shot distance of approximately 19.1–19.3 pitch units. The maximum recorded distance of 92.8 metres represents an extreme outlier corresponding to a shot from the shooter’s own half. It is retained in the dataset since it is a genuine attempt. Shot angle has a mean of 0.47 radians, with the maximum of 2.79 radians corresponding to shots taken from wider positions near the byline.

The tracking variables reveal important structural features of the data. The number of opponents in the freeze-frame ranges from a minimum of 3 to a maximum of 11, with a mean of approximately 9. This upper bound reflects the fact that freeze-frames capture only players visible in the camera frame rather than all outfield players. The cone opponents variable, measuring defenders directly obstructing the shooting lane, has a mean of 1.52 and a maximum of 8, indicating that most shots face at least one defender in the direct path to goal. The goalkeeper off-line distance is clipped at 20 metres following inspection of the raw data, which revealed three implausible values exceeding this threshold likely attributable to tagging errors or tracking noise.

Table 3.3: Descriptive statistics for key spatial and tracking variables.

Variable	Mean	Median	Std	Min	Max
Distance (m)	18.1	17.4	8.3	0.8	92.8
Angle (rad)	0.47	0.37	0.29	0.00	2.79
Number of Opponents	9.0	9.0	1.7	3	11
Cone Opponents	1.52	1.00	1.10	0	8
GK Off-Line Distance (m)	2.57	2.37	1.82	0.0	20.0
GK Distance (m)	13.1	11.9	8.2	0.0	40.6
Nearest Opponent Distance (m)	2.73	2.16	2.19	0.1	34.8

3.6 Body Part and Play Pattern Distribution

Table 3.4 presents the breakdown by body part. Right foot shots dominate at 51.7% of attempts, followed by left foot (29.1%) and headers (18.6%). The “Other” category, consisting of 45 shots using chest or other body parts, has the highest conversion rate at 22.2%, though the small sample size makes this figure unreliable. Headers convert at 10.8%, marginally higher than foot shots, which may appear counterintuitive but reflects the fact that headers predominantly occur from set pieces into high-value central areas. This distinction is examined in detail in Section 5.5.1, where the model’s differential performance on headers versus foot shots is analysed.

Table 3.4: Shot and goal counts by body part.

Body Part	Shots	Shots%	Goals	Conv.%
Right Foot	4,076	51.7%	363	8.9%
Left Foot	2,297	29.1%	225	9.8%
Head	1,467	18.6%	159	10.8%
Other	45	0.6%	10	22.2%
Total	7,885	100%	757	9.6%

Table 3.5 presents the play pattern distribution. Counter-attacks yield the highest conversion rate at 13.3%, as they take place during fast transitions of possession where the defensive structure is disorganised. Similarly, transitions originating from goalkeepers also convert well at 12.9%. Conversely, corners (8.5%) and free kicks (8.1%) convert below the overall aver-

age despite originating from set-piece situations, consistent with the well-organised defensive shape these situations allow. Regular play accounts for the largest share of shots (33.6%) at a conversion rate of 9.8%, close to the average. This pattern of counter-attacks converting at a substantially higher rate than other play patterns is consistent with Lucey et al. (2015), which similarly reports counter-attack conversion (14.87%) considerably exceeding open-play (8.26%) conversion rates.

Table 3.5: Shot and goal counts by play pattern.

Play Pattern	Shots	Goals	Conv.%
Regular Play	2,647	260	9.8%
From Throw In	1,545	156	10.1%
From Corner	1,439	122	8.5%
From Free Kick	1,333	108	8.1%
From Counter	354	47	13.3%
From Goal Kick	343	40	11.7%
From Keeper	132	17	12.9%
From Kick Off	92	7	7.6%
Total	7,885	757	9.6%

The difference between the relatively high conversion rate of headers (10.8%) and the below-average conversion of corners (8.5%) and free kicks (8.1%) may initially appear inconsistent, given that 63.3% of headers originate from set pieces. Intuitively, headers’ high conversion rate should be reflected in higher set piece conversion rates overall. In practice, set piece conversion is dragged down by the large volume of direct free kick attempts and foot shots taken from distance against organised defensive structures. Headers, by contrast, tend to arrive in high-value central areas, explaining their above-average conversion rate despite the crowded defensive structures characterised by set piece situations.

3.7 Descriptive Comparison: Goals vs Non-Goals

Table 3.6 presents mean feature values for goals and non-goals across all three feature categories, spatial, tracking, and temporal, providing initial descriptive evidence for the potential incremental value of each data stream in predicting shot outcomes.

Among spatial features, distance and angle show the largest differentials. Goals are taken from

32.4% closer range (12.6m vs 18.7m) and at 60.0% wider angles (0.707 vs 0.442 radians), in line with the findings of Rathke (2017) and Caley (2013). This pattern is consistent with Mead et al. (2023), which similarly finds that shot distance beyond approximately 20 metres is strongly associated with unsuccessful outcomes, and that successful shots cluster within a narrower angular range than unsuccessful attempts.

Among tracking features, goalkeeper angle error shows the strongest signal (0.262 vs 0.107 radians), indicating that shots taken when the goalkeeper is poorly positioned relative to the shooting trajectory are substantially more likely to result in goals, consistent with Anzer and Bauer (2021). Cone opponents are 23.0% lower for goals, consistent with the findings of Lucey et al. (2015) that fewer defenders obstructing the direct path to goal is associated with higher conversion rates.

Among temporal features, the through ball finding is notable. Shots preceded by a through ball convert at 25.4% versus 8.9% without. This provides descriptive motivation for the inclusion of possession sequence features in the modelling framework. Attack speed is 19.2% higher for goals (2.29 vs 1.92 m/s) and chain progressive distance 21.6% higher, suggesting that faster and more progressive attacks may generate higher quality chances. Counterintuitively, goals are preceded by 11.3% fewer defensive pressure events in the possession chain, suggesting that more direct attacks under less sustained pressure may generate higher quality shots, consistent with the counter-attack finding in Table 3.5.

Table 3.6: Mean feature values by outcome — spatial, tracking, and temporal features.

Feature	Non-Goal	Goal	Diff%	Type
Distance (m)	18.69	12.64	−32.4%	Spatial
Angle (rad)	0.442	0.707	+60.0%	Spatial
GK Off-Line (m)	2.491	3.315	+33.1%	Tracking
GK Angle Error (rad)	0.107	0.262	+144.9%	Tracking
Cone Opponents	1.555	1.197	−23.0%	Tracking
Attack Speed (m/s)	1.918	2.287	+19.2%	Temporal
Chain Prog. Distance (m)	39.5	48.0	+21.6%	Temporal
Chain Pressures	2.284	2.026	−11.3%	Temporal
Defensive Line X (m)	104.5	106.5	+2.0%	Temporal
Through Ball*	8.9%	25.4%	+185.4%	Temporal

3.8 Feature Correlation Structure

Figure 3.1 presents a lower-triangular correlation matrix for the key spatial and tracking features. Several patterns are worth noting. Distance and angle are strongly negatively correlated ($r = -0.756$), reflecting that shots from close range tend to subtend wider angles. GK distance is moderately correlated with shot distance ($r = 0.581$), as goalkeepers naturally position deeper when shots are taken from further out. Defenders in box and the number of opponents show moderate correlation ($r = 0.566$), indicating some redundancy between these two defensive density measures even though they capture distinct concepts (total visible opponents vs opponents inside the penalty area specifically). Notably, the goalkeeper off-line distance shows low correlation with most other features (maximum $|r| = 0.291$), suggesting it captures independent information about goalkeeper positioning not already encoded by distance or angle.

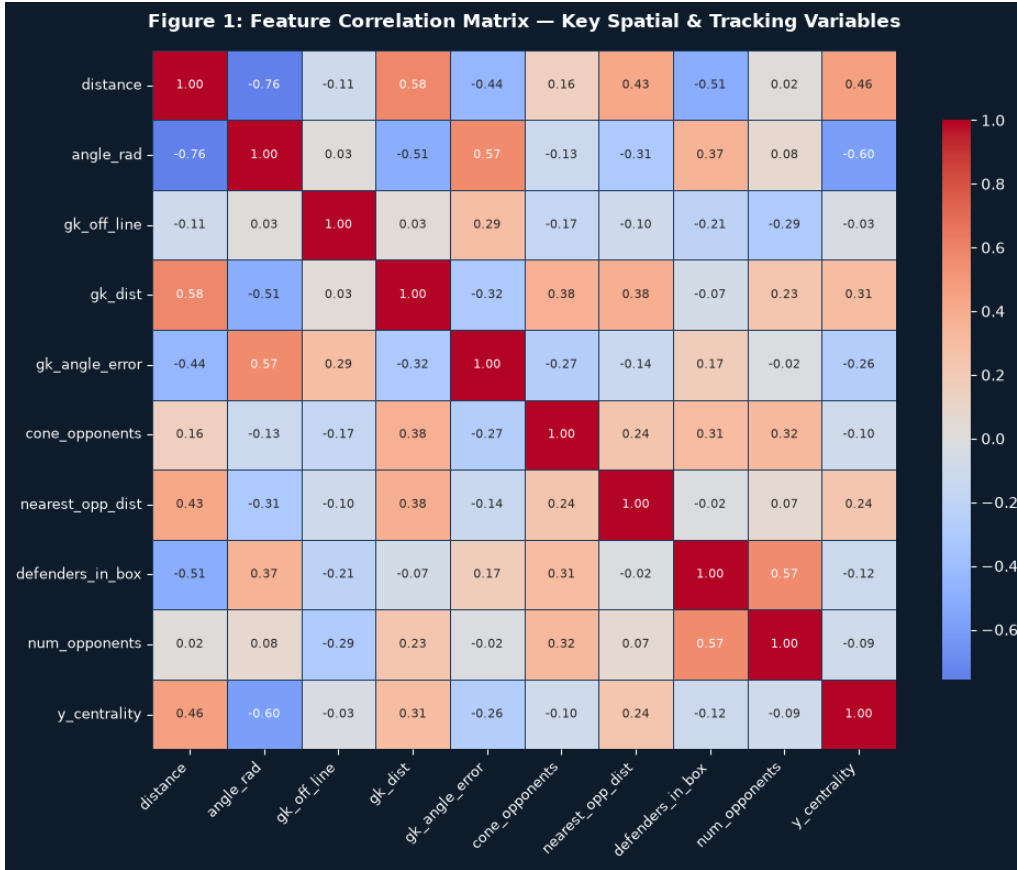


Figure 3.1: Feature correlation matrix for key spatial and tracking variables.

Figure 3.2 presents individual Pearson correlations with the binary goal outcome. Angle shows the strongest positive correlation ($r = 0.274$) and distance the strongest negative correlation ($r = -0.216$). Among tracking features, goalkeeper angle error ($r = 0.199$) and off-line distance ($r = 0.133$) show the strongest positive correlations, while defender in shot path ($r = -0.176$)

and goalkeeper distance ($r = -0.170$) show the strongest negative correlations. Among temporal features, through ball sequences ($r = 0.114$) show the strongest correlation with goal. This is higher than all tracking features except goalkeeper angle error and off-line distance. Defensive line ($r = 0.099$) and chain progressive distance ($r = 0.066$) also show meaningful positive correlations. The relatively modest individual correlations of tracking and temporal features do not imply their unimportance, as non-linear models can exploit interactions between features that univariate correlations cannot detect, discussed further in Section 4.2.6.

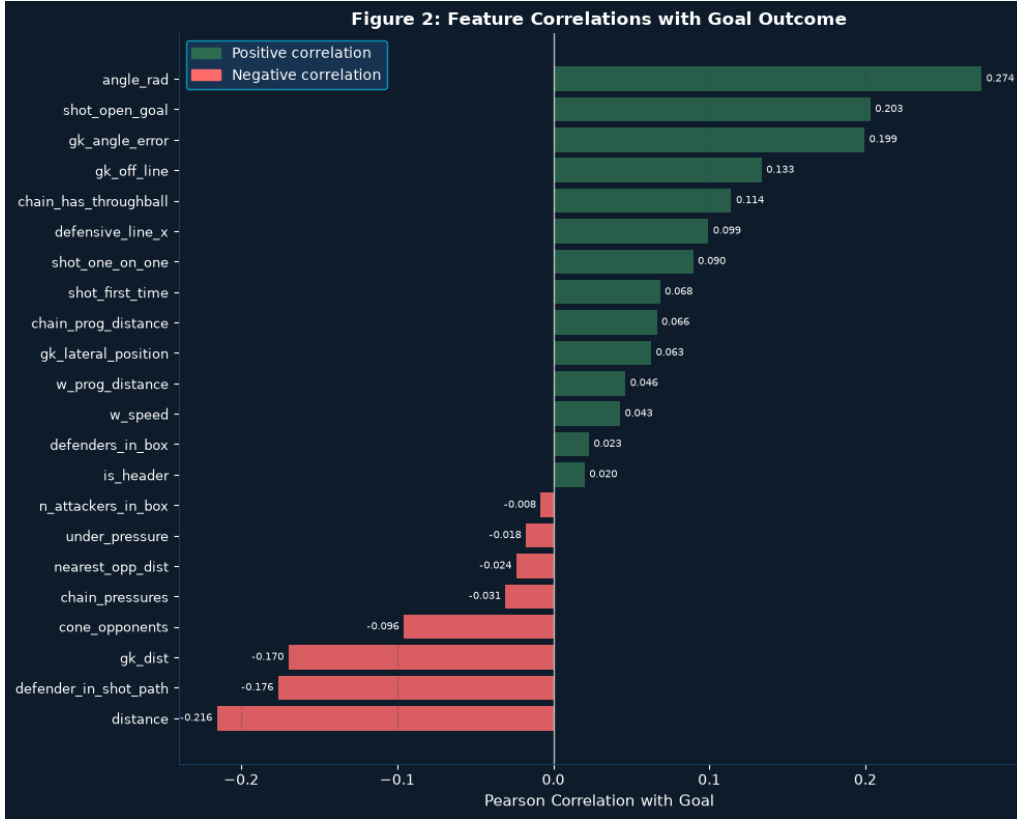


Figure 3.2: Feature correlations with goal outcome.

3.9 Shot Location Distribution

Figure 3.3 visualises the spatial distribution of all 7,885 shots, with white dots indicating goals. The hotspot is concentrated in the central area immediately in front of goal, roughly corresponding to the six-yard box and the area between the penalty spot and the goal line. A secondary concentration is visible around the penalty spot, consistent with the high volume of shots from this location in open play. The distribution has a long tail extending back toward the halfway line, reflecting the small number of long-range attempts in the dataset. Goals cluster centrally and close to goal, with relatively few goals from outside the penalty area.

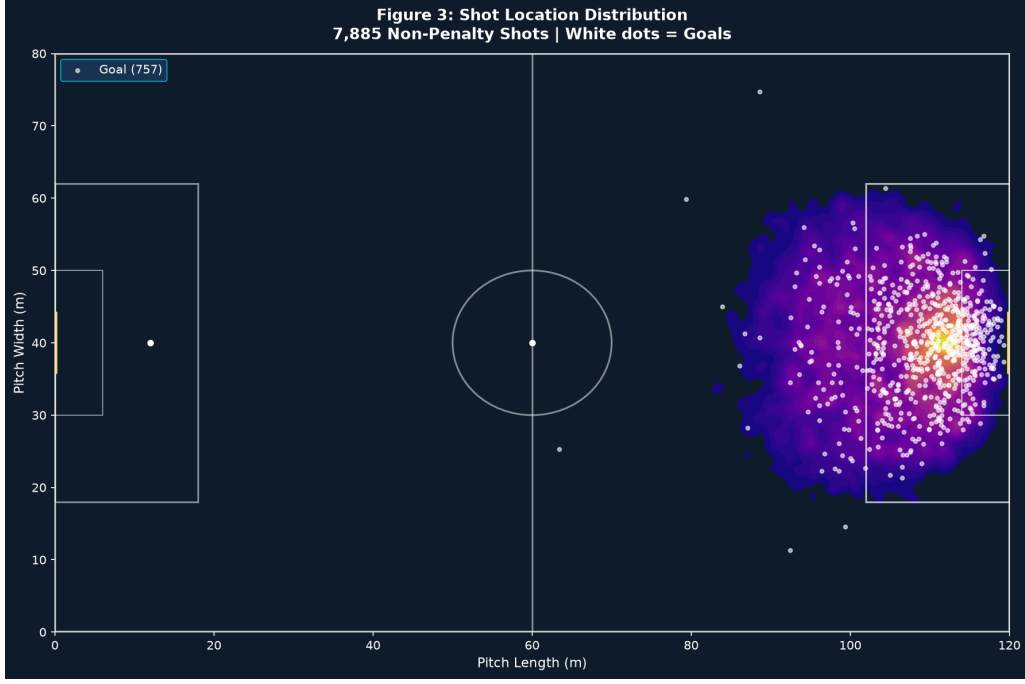


Figure 3.3: Shot location distribution across all 7,885 non-penalty shots.

3.10 Temporal Feature Overview

This study introduces additional temporal possession-chain features, along with the standard temporal features used in the literature. Full descriptive statistics for all temporal features are provided in Appendix A.

Table A2 (Appendix) shows that the mean possession chain contains 7.2 passes over 24.6 seconds, progressing 40.3 metres toward goal. However, chain duration ranges from 0 to 222 seconds, reflecting the wide variety of attacking sequences that culminate in shots, from fast breaks to prolonged possession phases. The median chain duration of 16.4 seconds is substantially below the mean, confirming right-skewness driven by a tail of long possession sequences. Attack speed has a mean of 1.95 m/s, with a maximum of 23.4 m/s reflecting fast break situations. The temporal features exhibit meaningful variation by competition context (Table A4, Appendix). Counter-attacks have the highest average attack speed (3.07 m/s) and fewest chain passes (3.5), consistent with their high conversion rate of 13.3%. Corners show distinctly low attack speed (0.58 m/s), reflecting the static nature of set-piece delivery. Through ball percentage is highest for counter-attacks (8.5%), confirming that fast transitions more frequently involve through balls that penetrate defensive lines.

Chapter 4

Methodology

4.1 Research Design

This study adopts an ablation-style experimental design to isolate the marginal contribution of tracking and temporal data to xG model performance. Rather than simply comparing a single model with and without tracking features, the design systematically evaluates several distinct machine learning model families across four feature conditions: a pure event-data baseline, a tracking-enhanced feature set, a temporal feature set, and a combined feature set. This allows assessment of the extent to which each data stream improves xG prediction, and which model family benefits most from their inclusion.

The experimental design follows the approach established by Anzer and Bauer (2021), which quantifies the value of tracking data through direct comparison against an event-data baseline on the same shots. This approach is extended by comparing across multiple model families rather than a single algorithm, consistent with Mead et al. (2023).

All models are trained and evaluated on identical train/test splits to ensure comparability. An 80/20 stratified split is used, with stratification on the binary goal outcome to preserve the natural class imbalance in both partitions. This follows standard practice for imbalanced classification tasks (Hastie et al., 2009), consistent with the stratified cross-validation approach explicitly adopted by Eggels (2016) and the match-level train/validation/test splitting used by Anzer and Bauer (2021) to avoid overrepresenting specific teams. The random seed is fixed to ensure full reproducibility.

4.2 Feature Engineering

Feature engineering is central to this study, as the core research question concerns the marginal contributions of spatiotemporal tracking-based features beyond standard event data. Following the demonstration by Hewitt and Karakuş (2023) that expanding the feature set of a logistic regression substantially improves its performance without increasing model complexity, this study invests heavily in constructing theoretically motivated features rather than relying on model flexibility alone. The full feature set of 82 variables is organised into six categories.

4.2.1 Baseline Spatial Features

The baseline feature set follows the foundational xG literature (Pollard et al., 2004; Rathke, 2017) and includes the two primary spatial determinants of shot conversion probability. Shot distance is computed as the Euclidean distance from the shot location to the centre of the goal line:

$$\text{distance} = \sqrt{(120 - x)^2 + (40 - y)^2} \quad (4.1)$$

Shot angle is computed as the angle subtended by the two goalposts from the shooter’s position:

$$\text{angle} = \arccos \frac{\vec{v}_1 \cdot \vec{v}_2}{|\vec{v}_1||\vec{v}_2|} \quad (4.2)$$

where $\vec{v}_1$ and $\vec{v}_2$ are vectors from the shot location to each goalpost.

Additional spatial features include y -centrality (deviation from the central axis of the goal), near-post distance, and far-post distance, which together capture the full geometric relationship between the shot location and the goal frame.

4.2.2 Shot Context Features

Event-data contextual features capture the situational characteristics of each shot. Binary flags are constructed for shots under pressure, first-time shots, one-on-one situations, and open goal situations, all sourced directly from StatsBomb event annotations. Body part is included as a feature, encoded as the historical mean conversion rate for each category via target encoding. Shot technique and play pattern are encoded as dummy variables following standard practice (Mead et al., 2023). Two additional binary flags identify aerial set-piece headers and diving

headers, motivated by the observation in this dataset that headers arise predominantly from set-piece deliveries into congested central areas (Section 3.6), and by evidence that headers constitute a structurally distinct shot type requiring separate modelling treatment (Anzer and Bauer, 2021).

4.2.3 Tracking-Derived Goalkeeper Features

Three features capture goalkeeper positioning at the moment of the shot, derived from StatsBomb 360 freeze-frame coordinates: goalkeeper distance, the Euclidean distance between goalkeeper and shooter; goalkeeper off-line distance, the goalkeeper’s distance from the goal line; and goalkeeper angle error, the angular deviation of the goalkeeper from the direct shooter-to-goal line, where a larger error indicates poorer positioning relative to the shooting trajectory. These features are motivated by evidence that goalkeeper positioning carries meaningful predictive information beyond shot geometry alone, as demonstrated by Anzer and Bauer (2021). Goalkeeper lateral position additionally captures horizontal deviation from goal centre, identifying positioning bias toward one side that may leave part of the goal exposed.

4.2.4 Tracking-Derived Defender Features

Six features capture defensive structure and pressure at the moment of the shot, operationalising the shooting lane obstruction concept introduced by Lucey et al. (2015). The total number of opponents in the freeze-frame provides a global measure of defensive presence, though this is bounded by camera coverage rather than the full outfield. Cone opponents counts defenders positioned within the geometric triangle from the shooter to both goalposts, directly implementing the defensive triangle descriptor. Nearest opponent distance measures the Euclidean distance from the shooter to the closest defender, capturing immediate physical pressure. Nearest block distance measures the perpendicular distance of the closest defender to the shot trajectory line, capturing obstruction of the shooting lane rather than proximity to the shooter. Defenders in the box counts opponents positioned within the penalty area, capturing the density of defensive coverage in the critical zone. Additionally, defender in shot path is a binary indicator for whether any defender lies within 1.5 metres of the direct line between the shooter and the goal centre, providing a stricter measure of direct shot obstruction than the cone measure. The number of attackers in the box and attackers ahead of the ball capture the attacking team’s numerical presence in the danger zone, while cone advantage (teammates minus opponents in the shot cone) captures the net numerical balance in the shooting lane.

4.2.5 Temporal Possession Features

Temporal features capture the dynamics of the attacking move rather than the static snapshot at the moment of the shot, capturing speed, structure, and defensive disorganisation. This complements the approach of Anzer and Bauer (2021) by incorporating the temporal dimension of possession sequences, following the event-sequence framework of Eggels et al. (2016).

Two sets of temporal features are constructed. The first captures the 10-second window immediately preceding the shot. Features in this window include the number of passes and carries completed, the number of pressure events and dribbles, the progressive distance covered, and the speed of the attack (progressive distance divided by duration). These window features capture the immediate momentum and intensity of the attack in the seconds before the shot.

The second set captures the full possession chain leading to the shot. Chain-level features include total passes and carries in the possession, chain duration, the number of pressure events faced, whether the chain included a cross, a through ball, or a switch of play, the starting x -coordinate of the possession, and the total progressive distance of the chain. Additional binary features identify whether the shot followed a rebound (block, save, or clearance in the preceding three events), whether the possession constitutes a fast break (chain duration under four seconds and attack speed above three metres per second), whether a pass or carry entered the penalty box during the chain, the number of defenders beaten by dribble, and the x -coordinate of the mean defensive line at the moment of the shot.

Critically, while pre-shot spatial dynamics are retained, the retrospective goal-assist indicator is excluded to prevent target leakage. Because this attribute is assigned conditionally on a goal being scored, its inclusion would directly encode the outcome variable, artificially inflating performance with information unobservable at the time of prediction.

4.2.6 Interaction Terms

Fourteen interaction terms are constructed to capture theoretically motivated joint effects between features that additive linear models cannot represent without explicit specification. These include six spatial interactions, for example goalkeeper distance $\times$ off-line distance, capturing combined goalkeeper commitment and distance, and distance $\times$ defenders in box, capturing compounded shot difficulty, alongside eight temporal interactions, for example attack speed $\times$ distance and through ball $\times$ angle. While these interactions are manually specified for the linear model, tree-based models can discover additional feature interactions automatically through se-

quential splitting, without requiring explicit specification (Chen and Guestrin, 2016). For this reason, the manually engineered interaction terms are included only in the feature sets used for logistic regression, while tree-based models are provided with the uninteracted feature set, allowing each model family to be evaluated using the feature representation best suited to its underlying mechanism.

4.3 Model Families

Five machine learning model families are evaluated, selected to span the range of complexity and flexibility relevant to tabular sports analytics data. This selection is motivated by the consistent finding in the literature that gradient boosting methods outperform alternatives on structured football data (Corsaro et al., 2025; Mead et al., 2023), while also including simpler approaches to test whether this advantage holds under tracking-enhanced features.

Logistic Regression (LR) serves as the interpretable linear baseline. Despite its simplicity, logistic regression has proven competitive in xG modelling when features are well-engineered (Hewitt and Karakuş, 2023). ElasticNet regularisation is applied combining L1 (LASSO) and L2 (Ridge) penalties, with optimal regularisation parameters identified via randomized search cross validation. Features are standardised prior to fitting, following standard practice for linear models (Noordman, 2019). The inclusion of interaction terms is particularly important for logistic regression as it cannot capture feature interactions automatically.

Random Forest (RF) is a bagging ensemble of decision trees that captures nonlinear feature interactions through bootstrap aggregation (Breiman, 2001). Random forest has been widely applied in football analytics (Baboota and Kaur, 2019; Mead et al., 2023) and serves as a nonlinear baseline that does not rely on sequential boosting.

XGBoost (XGB) is a regularised gradient boosting framework that builds trees sequentially, with each tree correcting the residual errors of the previous one (Chen and Guestrin, 2016). XGBoost consistently achieves strong performance on tabular football data in the literature (Anzer and Bauer, 2021; Mead et al., 2023) and is the primary tree-based model of interest in this study.

LightGBM is an alternative gradient boosting implementation using leaf-wise tree growth rather than depth-wise, which can yield faster training and different bias-variance tradeoffs on structured data (Corsaro et al., 2025). Its inclusion allows assessment of whether any XGBoost advantage is specific to that implementation or general to gradient boosting.

CatBoost (CB) is a third gradient boosting implementation that uses ordered boosting and symmetric trees, which has shown strong performance on sports analytics datasets with mixed feature types (Prokhorenkova et al., 2018; Noordman, 2019). Its inclusion completes the gradient boosting comparison and provides additional robustness.

Neural networks were also evaluated as an additional model family, following their application in comparable xG studies (Méndez et al., 2023; Corsaro et al., 2025). Neural networks are theoretically capable of learning arbitrarily complex nonlinear relationships without requiring manual feature engineering, and have been shown to achieve competitive performance in some xG applications on larger samples (Méndez et al., 2023). However, neural networks generally require substantially larger training sets than tree-based methods to realise this advantage on tabular data (Grinsztajn et al., 2022; Shwartz-Ziv and Armon, 2022), motivating their inclusion here as a point of comparison despite the comparatively modest sample size available in this study.

A stacking ensemble is additionally constructed using the five tuned models as base learners, with out-of-fold predictions fed to a logistic regression meta-learner (Wolpert, 1992).

4.4 Hyperparameter Tuning

Hyperparameter optimisation is performed for all model families using randomized search cross validation, optimising for AUC-ROC. This is preferred over exhaustive grid search cross validation following Bergstra and Bengio (2012), which demonstrates that random search finds equally good or better hyperparameter configurations in fewer evaluations when the parameter space is large.

Logistic regression is tuned over the regularisation strength C and the ElasticNet mixing parameter. Random forest is tuned over the number of estimators, maximum depth, minimum samples per leaf, minimum samples for splitting, maximum features per split, and the subsample ratio. XGBoost is tuned over nine hyperparameters including tree depth, learning rate, subsampling ratios, regularisation terms, and minimum child weight, with 30 iterations and 3-fold cross-validation. CatBoost is tuned over iterations, depth, learning rate, L2 regularisation, subsample ratio, and column subsample ratio. LightGBM is tuned over nine hyperparameters including number of leaves, tree depth, number of estimators, learning rate, minimum child samples, subsampling ratios, and regularisation terms.

Cross-validation folds and search iterations were scaled to the size of each model’s hyperparam-

eter space, ranging from 3 to 5 folds and 20 to 75 iterations across model families to manage computational cost. All hyperparameter searches are conducted exclusively on the training set to prevent information leakage from the test set into model selection decisions (Hastie et al., 2009). The optimal configurations identified are presented in Table 4.1.

Table 4.1: Optimal hyperparameter configurations identified by RandomizedSearchCV.

Model	Hyperparameter	Optimal Value
LR ElasticNet	C	0.05
	<code>l1_ratio</code>	1.0
Random Forest	<code>n_estimators</code>	300
	<code>max_depth</code>	None
	<code>min_samples_leaf</code>	10
	<code>min_samples_split</code>	5
	<code>max_features</code>	sqrt
	<code>max_samples</code>	0.7
XGBoost	<code>n_estimators</code>	700
	<code>max_depth</code>	5
	<code>learning_rate</code>	0.01
	<code>subsample</code>	0.7
	<code>colsample_bytree</code>	0.6
	<code>min_child_weight</code>	1
	<code>gamma</code>	0.3
	<code>reg_alpha</code>	0.5
	<code>reg_lambda</code>	2.0
CatBoost	<code>iterations</code>	300
	<code>depth</code>	4
	<code>learning_rate</code>	0.1
	<code>l2_leaf_reg</code>	10
	<code>subsample</code>	0.8
	<code>colsample_bylevel</code>	0.7
LightGBM	<code>num_leaves</code>	15
	<code>max_depth</code>	3
	<code>n_estimators</code>	500
	<code>learning_rate</code>	0.02
	<code>min_child_samples</code>	20
	<code>subsample</code>	1.0
	<code>colsample_bytree</code>	0.8
	<code>reg_alpha</code>	0.1
	<code>reg_lambda</code>	0.01

4.5 Evaluation Metrics

Model performance is evaluated using two complementary metrics. The primary metric is the Area Under the ROC Curve (AUC-ROC), which measures the probability that a randomly selected goal is assigned a higher xG than a randomly selected non-goal. AUC is threshold-independent and robust to class imbalance, making it the appropriate primary metric for xG evaluation (Fawcett, 2006). This choice is in line with the majority of the xG literature (Eggels, 2016; Mead et al., 2023; Noordman, 2019; Anzer and Bauer, 2021). Individual model AUC estimates are reported with 95% bootstrap confidence intervals (Efron and Tibshirani, 1993), computed via percentile bootstrap with 5,000 resampling iterations.

The secondary metric is Log Loss (binary cross-entropy), which directly penalises poorly calibrated probability estimates and is sensitive to the full probability distribution rather than just rank ordering. Log loss is particularly relevant for xG models because xG values are used as absolute probability estimates by practitioners, and is similarly reported across comparable xG studies (Eggels, 2016; Mead et al., 2023; Noordman, 2019). Calibration curves and precision-recall curves are additionally reported as complementary diagnostics, evaluating probability reliability across the prediction range and model performance under class imbalance respectively. Average precision (AP), the area under the precision-recall curve, is reported as a summary statistic for precision-recall performance, analogous to AUC for the ROC curve.

To assess whether observed differences in AUC between models are statistically significant rather than attributable to sampling variation, paired DeLong tests are conducted throughout the analysis (DeLong et al., 1988). The DeLong test is a non-parametric method for comparing the areas under two correlated ROC curves evaluated on the same test observations, exploiting the paired structure of the data to achieve greater statistical power than unpaired alternatives. The null hypothesis is $H_0 : \Delta\text{AUC} = 0$, tested against the two-sided alternative. Where multiple simultaneous comparisons are conducted, Holm-Bonferroni correction is applied to control the familywise error rate and prevent spurious significance in borderline cases (Holm, 1979).

Each shot in the dataset includes StatsBomb’s own xG estimate. These pre-computed xG values are used throughout this study as a benchmark against which the models developed here are evaluated. StatsBomb’s published methodology describes a suite of five gradient-boosted trees models, including separate pre-shot chance-quality and post-shot shot-execution specifications, trained on approximately 1.45 million shots and validated on a further 360,000 (StatsBomb, 2022). It also employs continuous spatial representations of defender and goalkeeper positions

rather than the discrete counts used here such as the number of defenders in the box, and applies monotonic constraints and goal-rate recalibration for shot-sparse regions, none of which are implemented in the present study.

Chapter 5

Results

5.1 Core Finding: The Marginal Contribution of Spatiotemporal Data

The central research question concerns whether spatiotemporal tracking data meaningfully improves xG prediction beyond standard event-data features. As a first step, the individual contributions of freeze-frame tracking features and temporal possession features are isolated in a controlled ablation using logistic regression, comparing a pure event-data baseline against tracking-enhanced and temporal-enhanced specifications separately.

Table 5.1 presents the result of this controlled comparison. Tracking features provide a larger lift than temporal features, both achieved with no change in model complexity, providing controlled estimates of the marginal predictive value of each feature category in isolation. These comparisons establish the direction and magnitude of each contribution before tuning. Statistical significance of performance gains across the full feature space and after hyperparameter tuning is examined in Section 5.2.

Table 5.1: Controlled ablation isolating the marginal contribution of freeze-frame tracking and temporal possession features using untuned logistic regression.

Model	AUC	Log Loss	Lift vs Baseline
StatsBomb	0.867	0.228	—
LR — With Tracking	0.841	0.243	+0.015
LR — With Temporal	0.832	0.246	+0.006
LR — No Tracking	0.826	0.254	—

Tracking features provide the larger marginal lift, consistent with evidence that the geometric configuration at the moment of the shot is a primary determinant of conversion probability (Anzer and Bauer, 2021). Temporal features provide a smaller but complementary contribution, capturing build-up context not reflected in the static freeze-frame snapshot.

This event-data-only baseline of 0.826 is broadly comparable to event-data specifications reported elsewhere in the literature (AUC 0.800, Mead et al., 2023; AUC 0.79, (Appiah et al., 2026); event-only baseline AUC 0.772, Anzer and Bauer, 2021), suggesting the present dataset and feature set are representative of standard event-based xG modelling. Anzer and Bauer (2021) reports a larger tracking-driven AUC improvement of 0.051 on a considerably larger dataset.

5.2 Full Model Leaderboard

Having established the baseline marginal contribution of spatiotemporal data in isolation, the analysis is extended to evaluate the full feature space on the tuned models. Neural networks did not outperform the other specifications in any feature condition and are therefore excluded from the main comparison, consistent with evidence that gradient boosting methods generally dominate neural networks on tabular data at moderate sample sizes (Grinsztajn et al., 2022; Shwartz-Ziv and Armon, 2022). Full results are provided in Appendix B. Table 5.2 presents the best tuned result per model family alongside StatsBomb, with 95% bootstrap confidence intervals reported for each AUC estimate (5,000 iterations, percentile method).

Table 5.2: Best tuned model per family on the test set, with 95% bootstrap confidence intervals. Efficiency is expressed as a percentage of StatsBomb AUC. 'Full' = T+T with categorical dummies.

Model	Feature Set	AUC	95% CI	Log Loss	Gap to SB	Efficiency
StatsBomb	—	0.867	[0.835, 0.894]	0.228	—	100.0%
LR Tuned	T+T	0.850	[0.817, 0.879]	0.243	0.017	98.0%
RF Tuned	T+T	0.846	[0.812, 0.877]	0.245	0.021	97.6%
XGB Tuned	Full	0.835	[0.799, 0.867]	0.246	0.032	96.3%
CB Tuned	Full	0.830	[0.796, 0.862]	0.248	0.037	95.8%
LGB Tuned	Full	0.830	[0.795, 0.863]	0.248	0.037	95.8%

To isolate the contribution of feature engineering independently of hyperparameter tuning, Table 5.3 compares untuned baseline models against untuned T+T specifications for each model

family. Since both models are untuned, any AUC difference is attributable purely to the addition of tracking and temporal features. Paired DeLong tests with Holm-Bonferroni correction are reported for each comparison.

Table 5.3: Feature engineering contribution: untuned baseline vs untuned T+T specification per model family. Δ AUC reflects the pure effect of adding tracking and temporal features, with tuning held constant. Significance stars based on Holm-corrected p-values from paired DeLong tests across five simultaneous comparisons.

Model	Baseline AUC	T+T AUC	Δ AUC	Holm p
LR	0.826	0.832	+0.006	0.660
RF	0.791	0.842	+0.051***	0.001
XGB	0.782	0.816	+0.034	0.067
CB	0.796	0.838	+0.042**	0.010
LGB	0.740	0.801	+0.061**	0.009

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Several findings from Tables 5.2 and 5.3 warrant discussion.

Tracking data consistently outperforms temporal features as a source of predictive lift. Across all five model families, the tracking feature set outperforms the temporal feature set in isolation. Tracking lifts range from +0.015 (LR) to +0.064 (LGB), while temporal lifts range from +0.006 (LR) to +0.054 (LGB). However, temporal features provide meaningful complementary signal, particularly for tree-based models where the combined T+T specification consistently outperforms either feature set alone.

Performance gains are statistically robust across all model families. Paired DeLong tests confirm that the AUC improvement from untuned baseline to final tuned specification is statistically significant for all five model families after Holm correction (all Holm-corrected $p < 0.05$), with Δ AUC ranging from +0.024 (LR) to +0.090 (LGB). These combined gains capture the joint effect of feature engineering and hyperparameter tuning. To isolate the pure feature engineering contribution, Table 5.3 shows that RF, CatBoost, and LightGBM show statistically significant AUC gains from adding tracking and temporal features alone, before any tuning is applied.

Logistic regression with the complete feature set is the best overall model. The tuned ElasticNet logistic regression with combined tracking and temporal features achieves the

highest AUC and lowest log loss, representing 98.0% of StatsBomb’s discriminative performance. Paired DeLong testing (Appendix B) reveals that LR Tuned and RF Tuned are the only models whose gaps to StatsBomb do not reach statistical significance after Holm correction (both Holm $p = 0.095$), indicating near-commercial performance for both linear and ensemble tree specifications. By contrast, XGBoost, CatBoost, and LightGBM all differ significantly from StatsBomb (Holm $p \leq 0.09$). This demonstrates that a well-engineered linear model achieves near-commercial performance, echoing the finding of Hewitt and Karakuş (2023) that careful feature engineering can substitute for model complexity at small sample sizes. A stacking ensemble across all five families matches this AUC but with higher log loss (0.250), offering no meaningful improvement over the best single model.

Feature engineering benefits tree-based models significantly more than logistic regression. The statistically insignificant feature engineering gain for LR ($\Delta\text{AUC} = +0.006$, Holm $p = 0.660$) reflects that its baseline is already the strongest of any model family (0.826), because the core spatial features are largely linear in their relationship with goal probability, which logistic regression exploits effectively even without additional features. The full gain for LR ($\Delta\text{AUC} = +0.024$) is only realised after hyperparameter tuning, when the ElasticNet specification selects interaction terms that capture more complex relationships between tracking and temporal features. Tree-based models, by contrast, show substantially larger relative gains from feature engineering alone, suggesting they are better equipped to exploit such interactions (Chen and Guestrin, 2016).

LightGBM records the largest absolute gain but consistently underperforms relative to other model families. Despite a large combined ΔAUC (+0.09, Holm $p < 0.001$), tuned LightGBM (0.830) remains below tuned RF (0.846) and XGBoost (0.835) and its gap to StatsBomb is statistically significant (Holm $p = 0.004$). Even with a moderately complex configuration, LightGBM’s leaf-wise growth strategy appears to interact unfavourably with the moderate feature correlations in the tracking feature set at this sample size (Hastie et al., 2009), leaving it unable to match RF or XGBoost’s depth-wise tree construction. LightGBM’s large absolute gain therefore reflects its weak baseline (0.740) rather than superior feature exploitation.

Positioning within the wider literature. Table 5.4 situates the present study’s results against comparable expected goals models reported elsewhere. This comparison should be interpreted cautiously, since differences in competition coverage, data provider, time period, sample size, and shot filtering criteria across studies substantially limit direct comparability.

The best model in this study achieves the highest AUC among all comparable studies listed, exceeding Anzer and Bauer (2021) despite a moderately smaller sample size and instantaneous freeze-frame snapshots rather than continuous tracking. Whether this reflects genuine methodological strength, favourable characteristics of the specific competitions included, or differences in evaluation procedure across studies cannot be fully determined from this comparison alone.

Table 5.4: Indicative comparison to prior xG literature. AUC values are not directly comparable across studies due to differences in data source, competition coverage, and sample size. n.r. = not reported in the source.

Study	Best Model	Data Type	Sample	AUC
StatsBomb (benchmark)	Proprietary	Tracking	7,885	0.867
This study	LR (tuned, T+T)	Tracking + Temporal	7,885	0.850
Anzer and Bauer, 2021	Gradient Boosting	Tracking (25 Hz)	105,627	0.823
Eggels et al., 2016	Random Forest	Event only	128,667	0.814
Noordman, 2019	CatBoost	Event + player qual.	146,540	0.802
Mead et al., 2023	XGBoost	Event + contextual	—	0.800
(Appiah et al., 2026)	Mixed-effects LR	Event only	10,709	0.790
Eggels, 2016	RF (tuned)	Tracking (10 Hz)	n.r.	0.775

5.3 SHAP Feature Importance Analysis

5.3.1 Cross-Scenario SHAP Analysis

To understand which features drive model predictions and how the importance structure shifts as richer feature sets are introduced, SHAP values are computed for five model specifications (Lundberg and Lee, 2017). Five scenarios are evaluated, corresponding to the four untuned feature conditions plus the best overall tuned model. Random Forest is selected as the representative tree-based model for Scenarios 1 through 4, as it constructs trees independently through bagging rather than the sequential, hyperparameter-sensitive boosting process, providing a more stable basis for isolating the effect of feature category changes (Hastie et al., 2009). Scenario 5 switches to the best overall model, ElasticNet LR, to examine whether model family affects the importance structure when the same combined feature set is used.

Table 5.5 summarises the percentage of total SHAP importance attributed to each feature category across all five scenarios. Figure 5.1 presents the feature type contribution evolution across scenarios. Full SHAP tables for all five scenarios are provided in Appendix A.

Table 5.5: SHAP importance decomposition by feature type across scenarios.

Scenario	Model	AUC	Baseline (%)	Tracking (%)	Temporal (%)
S1: Baseline	RF	0.791	100.0	0.0	0.0
S2: Tracking	RF	0.840	29.7	70.3	0.0
S3: Temporal	RF	0.834	48.7	0.0	51.3
S4: T+T	RF	0.842	23.1	51.3	25.6
S5: T+T	LR	0.850	27.7	23.7	48.6

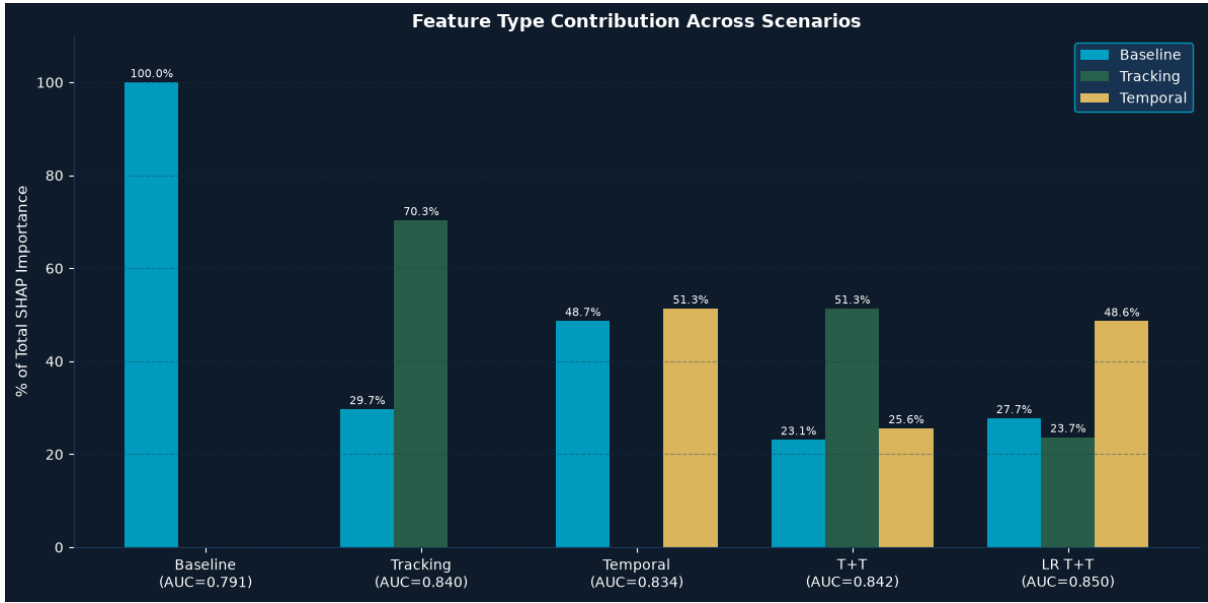


Figure 5.1: Feature type contribution evolution across scenarios.

Scenario 1 — Baseline Model (AUC = 0.791)

In the baseline specification, shot angle accounts for 36.7% of total importance, followed by distance (26.1%) and body part (14.2%), together explaining 77% of the model’s predictive power. This is broadly consistent with Mead et al. (2023), which similarly finds distance dominant in the Premier League specification. Hewitt and Karakuş (2023) finds comparable results in a baseline SHAP analysis, but concludes that overall SHAP magnitudes remain modest.

Scenario 2 — Adding Tracking Features (AUC = 0.840)

The addition of tracking data fundamentally shifts the importance structure. Tracking features collectively account for 70.3% of total SHAP importance, with baseline features declining to 29.7%, the largest structural shift across the five scenarios. Within tracking features, goalkeeper positioning dominates, with goalkeeper off-line distance and goalkeeper distance to goal ranking

highly alongside post distance proxies and nearest opponent distance. A notable finding is the prominence of the new feature defender in shot path, which ranks 6th overall (7.5%), ahead of established features such as goalkeeper angle error, suggesting a binary indicator for direct shot path obstruction captures distinct information from the continuous cone measure.

This parallels Anzer and Bauer (2021), whose SHAP analysis similarly identifies goalkeeper distance as the second most important feature after shot location, with a defensive pressure metric ranking third, consistent with Hewitt and Karakuş (2023) which finds comparable shifts in dominant contributors when goalkeeper and defensive features are added. Notably, shot angle remains the most important single feature (14.3%) even after adding 18 tracking features, though its share drops by more than half from the baseline, suggesting tracking features partially subsume the geometric information captured by shot angle.

Scenario 3 — Adding Temporal Features (AUC = 0.834)

When temporal features are added in place of tracking data, they account for 51.3% of model importance, broadly competitive with tracking’s 70.3% in Scenario 2. The most informative temporal feature is through ball in chain (5.4%), consistent with the descriptive finding in Section 3.7. Attack tempo features weighted speed (4.9%) and weighted progressive distance (4.0%) also rank highly. The lower AUC relative to Scenario 2 suggests temporal features provide marginally noisier signal than freeze-frame spatial data for tree-based models. This scenario suggests the added predictive value of the temporal possession features introduced in this study, extending SHAP-based feature attribution beyond the static freeze-frame context examined in prior work.

Scenario 4 — Tracking + Temporal (AUC = 0.842)

The combined model reveals a clear hierarchy for tree-based specifications. Tracking dominates at 51.3%, temporal contributes 25.6%, and baseline declines to 23.1%. The top temporal features all appear after the major tracking features, consistent with the AUC evidence in Table 5.3, where tracking alone outperforms temporal alone for RF. This ordering suggests that once goalkeeper positioning and defensive pressure are spatially accounted for, additional temporal context adds limited marginal information for Random Forest, explaining the modest combined improvement over tracking alone (+0.002 AUC).

Scenario 5 — ElasticNet LR Tracking+Temporal (AUC = 0.850)

Figure 5.2 presents the SHAP beeswarm plot for the best overall model, revealing a fundamentally different importance structure from the Random Forest specifications. In contrast to

Scenario 4 where tracking dominated, temporal features now account for 48.6% of total SHAP importance, while tracking contributes only 23.7%, reflecting the manually specified interaction terms in the LR specification that capture relationships between tracking and temporal features explicitly, whereas RF captures these implicitly through tree splits.

A striking finding is that the defensive line distance interaction term is the single most important feature at 23.8%, nearly double shot angle, which retains second place. This interaction term captures the compounded difficulty of shots taken from distance against a deep defensive line, a relationship potentially not captured by either feature in isolation. Nearest opponent distance, body part and cumulative chain pressure follow. The prominence of cumulative chain pressure ahead of goalkeeper off-line distance is a novel finding suggesting defensive pressure over the full possession carries meaningful signal in a linear specification, extending the pressure concept examined at the shot moment by Anzer and Bauer (2021) to the full build-up phase. The unobstructed angle feature ranks 9th, pointing to the added predictive value of the new spatial features introduced in this study.

This divergence between model families has practical implications for xG model design. The choice of model family should be informed by which feature type is theoretically prioritised.

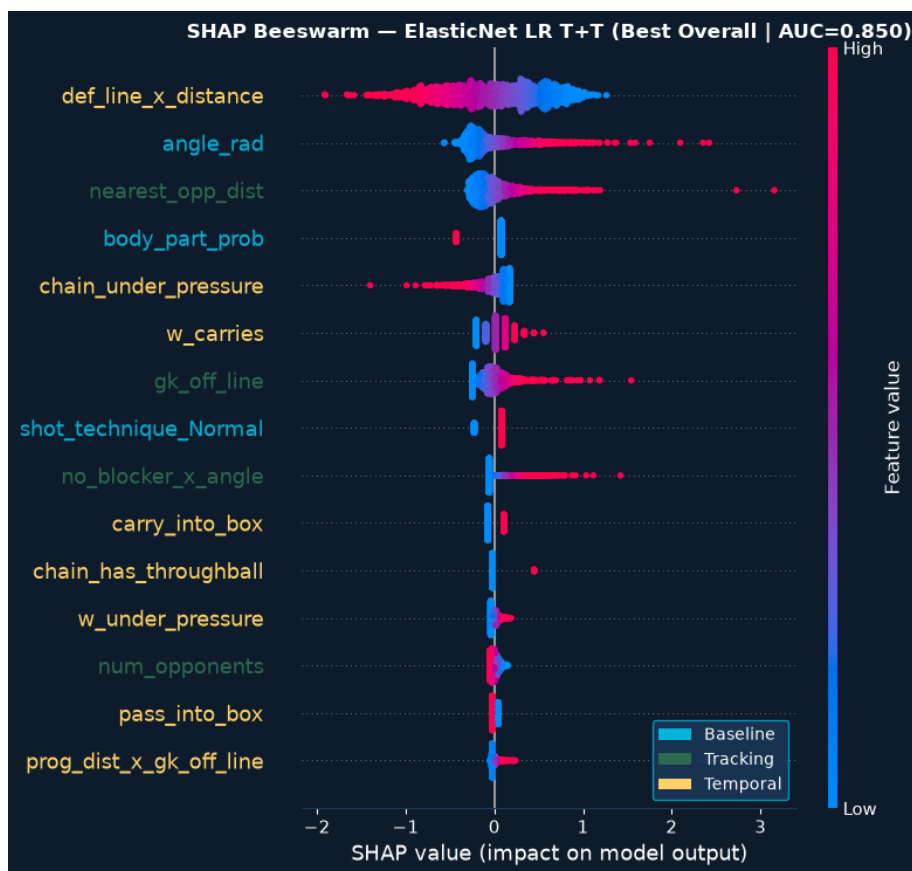


Figure 5.2: SHAP beeswarm plot for the best overall model (S5: ElasticNet LR).

5.3.2 ElasticNet Logistic Regression Coefficient Analysis

Table 5.6 presents key retained coefficient estimates for the best overall model. The ElasticNet penalty, equivalent to pure Lasso under strong regularisation, zeroes out 54 of 82 features, retaining 28 non-zero coefficients. Bootstrap confidence intervals (500 iterations) are computed to assess statistical significance, with 25 of 28 retained features significant at the 95% level. Of these, 7 are baseline, 8 are tracking, and 10 are temporal, indicating that all three feature types contribute statistically robust predictive signal to the best model. Coefficients are interpreted as odds ratios (OR), computed as $\exp(\beta)$.

Table 5.6: ElasticNet Logistic Regression: key retained features with bootstrap confidence intervals (500 iterations), sorted by absolute coefficient magnitude.

Feature	Category	Coefficient	95% CI
Defensive line $\times$ distance	Temporal	−0.564	[−0.648, 0.000]
Shot angle	Baseline	+0.332***	[0.182, 0.523]
Nearest opponent distance	Tracking	+0.259***	[0.115, 0.350]
Body part probability	Baseline	−0.196***	[−0.290, −0.001]
Goalkeeper off-line distance	Tracking	+0.175***	[0.097, 0.308]
Cumulative chain pressure	Temporal	−0.175***	[−0.319, −0.017]
Unobstructed angle	Tracking	+0.162***	[0.040, 0.248]
Ball carries (window)	Temporal	+0.149***	[0.001, 0.237]
Carry into box	Temporal	+0.087	[0.000, 0.190]
Number of opponents	Tracking	−0.043	[−0.147, 0.000]
Weighted pressure events	Temporal	+0.041	[0.000, 0.136]

*** Bootstrap 95% CI excludes zero on the log-odds scale (500 iterations).

Coefficients are log-odds. Features without stars have CIs touching zero due to Lasso regularisation in some bootstrap samples, rather than genuine uncertainty about the direction of the effect. Selected key features shown; 54 zeroed features omitted.

Positive Coefficients

Shot angle (OR = 1.393) is the strongest positive predictor, consistent with the foundational xG literature (Rathke, 2017). Nearest opponent distance (OR = 1.295) is the strongest tracking feature, with greater distance from the nearest opponent associated with a 29.5% increase in goal odds. The unobstructed angle feature (OR = 1.176) is the strongest new spatial feature, suggesting the combination of a wide shooting angle and an unobstructed shot path creates disproportionate danger. Ball carries in the build-up window is the strongest temporal feature,

with each additional carry associated with a 16.1% increase in goal odds, suggesting progressive ball-carrying generates higher quality shots. Carry into box ($OR = 1.091$) and weighted pressure events ($OR = 1.041$) also enter with positive coefficients, the latter suggesting that pressing situations may create fast transitions and disorganised defences rather than genuinely harder shots.

Negative Coefficients

The defensive line distance interaction ($OR = 0.569$) is the strongest negative predictor overall, reducing goal odds by 43.1%, consistent with its dominant SHAP contribution in Section 5.3.1. Cumulative chain pressure ($OR = 0.840$) reduces goal probability by 16.0%, extending the pressure concept from the shot moment to the full build-up phase. Body part probability ($OR = 0.822$) reflects the residual technical difficulty of headers once positional advantages are controlled for. Number of opponents ($OR = 0.958$) reduces goal probability by 4.2% per additional defender.

ElasticNet Feature Selection

The 54 features zeroed by Lasso provide insight into feature redundancy. Several temporal features including weighted speed, chain progressive distance, and fast break indicator are eliminated, suggesting collinearity with retained features or low predictive signal. Most play pattern dummies are zeroed, indicating that play pattern variation is adequately captured by the retained spatial and temporal features. The elimination of cone opponents and defenders in box suggests partial collinearity with nearest opponent distance and number of opponents respectively, an inherent limitation of freeze-frame tracking features given their mutual dependence on shot location and defensive organisation.

Coefficient Validation and Novel Findings

Coefficient signs are broadly consistent with existing xG studies. Rathke (2017) report angle and distance as dominant predictors, corroborated here with shot angle as the strongest positive predictor. Anzer and Bauer (2021) identify goalkeeper positioning as a key freeze-frame predictor, corroborated by the goalkeeper off-line distance coefficient. In contrast, (Appiah et al., 2026) find angular deviation, measured from the central axis rather than the angle subtended by the goalposts, insignificant once distance is controlled for in a bivariate specification without interaction terms ($\beta = -0.02$, $p = .69$). This divergence is likely attributable to the differing angle definitions, as well as the present study’s richer feature set and angle-based interaction terms, which may allow the model to isolate angle’s independent contribution more precisely. The temporal coefficients including ball carries, through ball in chain, and cumulative chain

pressure are, to the author’s knowledge, among the first bootstrap-validated coefficient estimates for possession-sequence features reported in the xG literature, supporting the inclusion of sequence-based features in future xG model specifications.

5.4 Model Diagnostics

5.4.1 ROC Analysis

Figure 5.3 presents the ROC curves for the three models alongside StatsBomb on the test set. All three models substantially outperform the diagonal random classifier baseline, indicating discriminative ability. The tuned LR tracks StatsBomb closely throughout the majority of the operating range, with the gap most pronounced in the low False Positive Rate ($\text{FPR} < 0.1$), where StatsBomb achieves a mean True Positive Rate (TPR) of 0.361 compared to 0.329 for LR and 0.316 for RF. This region consists of shots the model is most certain will result in goals, where StatsBomb’s proprietary model retains a meaningful advantage, likely reflecting access to additional data sources or proprietary features beyond the open freeze-frame data used here. XGBoost shows the largest gap at low FPR (mean $\text{TPR} = 0.280$), suggesting its high-confidence predictions are less precise than LR despite similar overall discrimination.

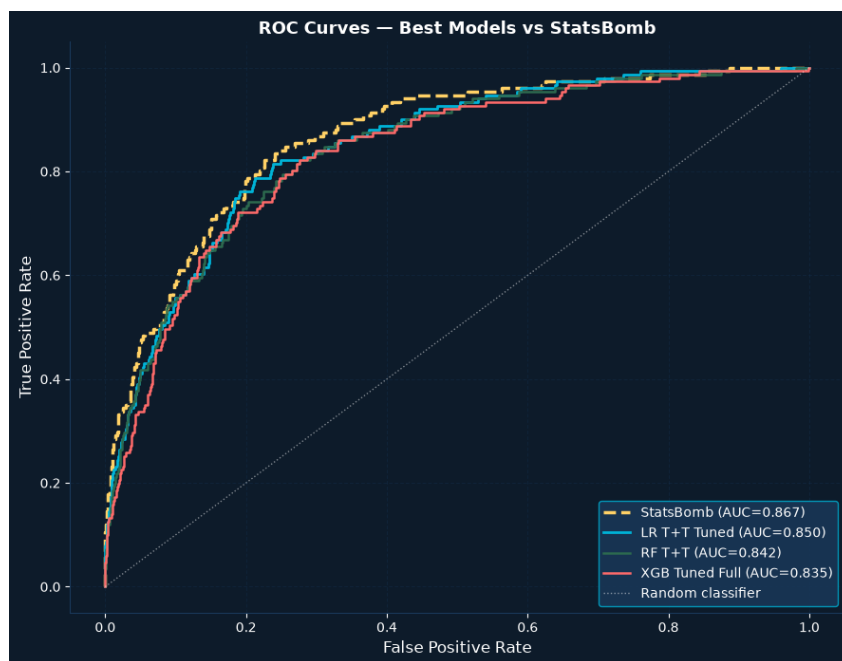


Figure 5.3: ROC Curves — Best Models vs StatsBomb.

5.4.2 Calibration Analysis

Figure 5.4 presents the calibration curves for all models. A notable finding is the strong mean-level calibration across all models. Both StatsBomb and LR predict a mean of 0.094 goals, RF 0.095, and XGBoost 0.090, against an actual conversion rate of 0.096. This calibration indicates that all models correctly estimate the overall goal rate without systematic misprediction at the aggregate level.

At the individual prediction level, all models are well-calibrated at low predicted xG values (0 to 0.3), which covers the large majority of shots. At higher predicted xG values the calibration curves become less stable, particularly for XGBoost and RF, driven by the small number of shots assigned high xG values rather than systematic miscalibration. StatsBomb’s curve is smoother throughout, consistent with the application of post-hoc calibration techniques such as the explicit goal-rate recalibration for shot-sparse regions described in StatsBomb’s published methodology (StatsBomb, 2022).

The practical implication is that while all models discriminate well between goals and non-goals, their absolute probability estimates at very high xG values (above 0.4) should be interpreted with caution. For applications requiring well-calibrated probabilities across the full range, isotonic regression calibration would be a natural refinement, particularly for XGBoost which shows the most pronounced instability at high predicted values, consistent with the calibration approach of Eggels (2016).

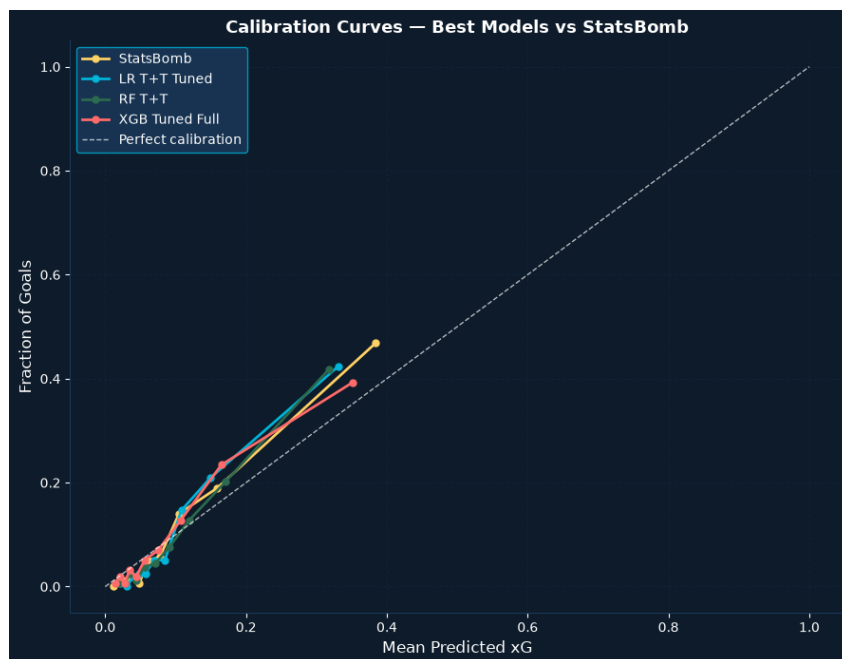


Figure 5.4: Calibration Curves — Best Models vs StatsBomb.

5.4.3 Precision-Recall Analysis

The Precision-Recall (PR) curves provide complementary diagnostic information under class imbalance. StatsBomb achieves the highest Average Precision (AP = 0.508), followed by LR, RF, and XGBoost. The ordering mirrors the AUC ranking, suggesting that the relative model performance is consistent across both evaluation frameworks. All models substantially outperform the random classifier baseline (AP = 0.096).

The gap between StatsBomb and LR is proportionally larger in the PR space than in ROC space, reflecting that StatsBomb’s advantage is concentrated in the high-precision, low-recall region. This corresponds to high-confidence predictions where its proprietary features provide the largest marginal benefit. This finding is consistent with the ROC gap at low FPR documented above and suggests that the remaining performance gap between open-data models and commercial providers is primarily in identifying the highest-quality chances rather than in overall discrimination.

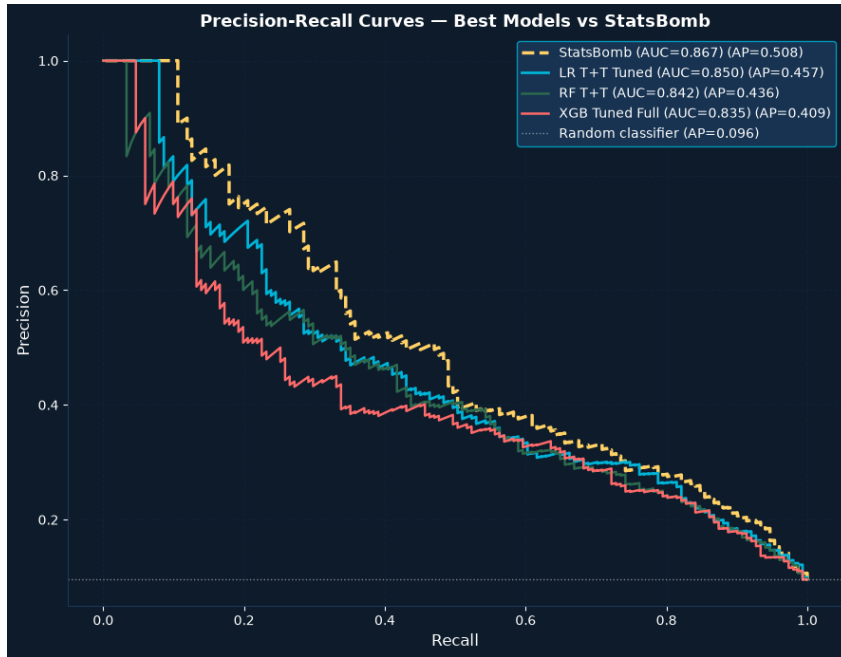


Figure 5.5: Precision-Recall Curves — Best Models vs StatsBomb.

5.5 Performance Breakdown

5.5.1 Header Gap Analysis

A key diagnostic finding concerns the model’s differential performance across shot types. Table 5.7 presents AUC scores separately for foot shots and headers on the test set across all

top-performing specifications, using StatsBomb as the benchmark.

Table 5.7: AUC performance breakdown by shot type across models on the test set.

Model	Foot AUC	Header AUC	Gap (Δ)	Gap to SB (Foot)	Gap to SB (Head)
StatsBomb	0.878	0.815	-0.063	—	—
LR T+T Tuned	0.862	0.803	-0.059	+0.016	+0.012
RF T+T	0.859	0.773	-0.086	+0.019	+0.042
XGB Tuned Full	0.847	0.778	-0.069	+0.030	+0.036

Test set composition: 1,267 foot shots (80.3%, 8.8% goal rate), 310 headers (19.7%, 12.6% goal rate).

Foot Shots

On foot shots, the LR model achieves a gap similar to the overall test set gap, suggesting that foot shots drive aggregate model performance. This indicates that freeze-frame features effectively capture the spatial and defensive context relevant to foot shot conversion, consistent with Lucey et al. (2015), which demonstrates that spatial defensive obstruction geometry is the primary driver of shot quality variation in open-play situations.

Headers

On headers, performance is highly model-dependent. The LR model achieves a gap comparable to its foot shot gap, indicating that the linear combination of freeze-frame spatial features and temporal build-up metrics captures header context as effectively as foot shot context relative to StatsBomb.

Tree-based models exhibit a distinct drop in aerial performance. Random Forest’s header AUC falls 0.030 below LR’s, a gap that does not exist on foot shots where both models perform comparably. This is consistent with SHAP findings in Section 5.3.1. Tree models rely heavily on freeze-frame tracking snapshots, whereas LR weighs temporal features more heavily. For headers, possession build-up dynamics carry predictive signal relative to a static spatial snapshot, explaining LR’s relative resilience in aerial evaluation.

Structural Characteristics & Limitations

Headers represent a structurally distinct event class, characterised by closer spatial proximity, wider angles, slower attack velocity, shorter possession chains, and distinct set-piece dynamics (Section 3.6). Subset SHAP analysis reveals that a defender in the shot path ranks 3rd in importance for headers, outranking post distance and goalkeeper positioning. This indicates that direct spatial obstruction is especially critical in aerial duels where shooting mechanics are

constrained.

Crucially, dynamic information absent from 2D freeze-frames, such as jump timing, aerial contest outcomes, body orientation, and ball trajectory, remains an unobserved gap relative to commercial models. Open 360 freeze-frame data lacks z -coordinate spatial tracking entirely, meaning aerial evaluation faces a structural data limitation that feature engineering can only partially mitigate.

5.5.2 Competition and Gender Subgroup Analysis

To evaluate model consistency across competitive contexts, AUC is evaluated separately by competition, gender, and tournament type on the test set. Table 5.8 summarises competition-level metrics for the three leading specifications alongside StatsBomb.

Table 5.8: Model AUC performance across competition subgroups on the test set.

Competition	n	Conv. %	StatsBomb	LR	RF	XGB	Best Gap to SB
Bundesliga (2023/24)	153	11.1%	0.888	0.845	0.863	0.864	+0.024 (XGB)
FIFA World Cup	267	10.5%	0.863	0.839	0.847	0.814	+0.016 (RF)
UEFA Euro	514	8.6%	0.850	0.855	0.822	0.820	-0.005 (LR)
UEFA Women’s Euro	326	9.5%	0.858	0.836	0.835	0.840	+0.018 (XGB)
Women’s World Cup	317	9.8%	0.888	0.872	0.869	0.854	+0.015 (LR)

Competition-Level Findings.

Performance varies across tournaments, with benchmark StatsBomb AUC ranging from 0.850 to 0.888. Notably, LR marginally exceeds StatsBomb on UEFA Euro. Given the sample size, this minor margin falls within expected sampling variance and should be interpreted as approximate parity rather than definitive superiority.

The Bundesliga exhibits the largest absolute gap to StatsBomb, likely reflecting the limited training volume available for domestic league play in the open release relative to international tournaments. Tree-based architectures outperform LR in the Bundesliga, though given the limited sample size this finding should be interpreted cautiously. Both the FIFA World Cup and Women’s World Cup display narrow performance gaps, suggesting reasonable cross-tournament generalisation.

At the aggregate level, performance across gender categories remains stable, with small gap variation in both cases, indicating that open-data feature engineering generalises across gender

categories.

Table 5.9: Granular AUC breakdown by gender and shot type interaction.

Subgroup	n	StatsBomb	LR T+T	Gap (Δ)
Men — All	934	0.863	0.848	+0.015
Women — All	643	0.874	0.853	+0.021
Men — Foot	756	0.880	0.875	+0.004
Men — Header	178	0.784	0.745	+0.039
Women — Foot	511	0.875	0.844	+0.030
Women — Header	132	0.852	0.874	-0.022

Gender $\times$ Shot Type Interaction.

Disaggregating by gender and shot type (Table 5.9) reveals notable granular patterns. On men’s foot shots, LR virtually matches commercial performance, the narrowest gap observed across all subgroups. Conversely, on men’s headers, the gap widens considerably, replicating the header attenuation noted in Section 5.5.1.

On women’s headers, LR outperforms StatsBomb. Given the limited subgroup size, this difference should be viewed as exploratory rather than conclusive. A plausible structural factor is that aerial chances in the women’s dataset originate predominantly from wide crosses with structured defensive block setups, which aligns well with explicit spatial angle and temporal sequence features. However, validating this hypothesis requires larger sample sizes in future open releases.

Overall, subgroup diagnostics indicate that while broad AUC metrics are robust, performance varies across shot mechanics and tournament contexts, with open linear-temporal models performing closest to commercial benchmarks on international foot shots.

5.6 Bundesliga Betting Case Study: Poisson Aggregation Framework

As an applied illustration of how shot-level xG estimates can be translated into match-level outcome probabilities, this section implements a Poisson aggregation framework following Dixon and Coles (1997) and evaluates its calibration relative to betting market prices. Shot-level xG predictions are generated using the tuned ElasticNet logistic regression model, the best overall model identified in Section 5.2. The analysis uses all 34 Bundesliga 2023/24 matches with

StatsBomb 360 data, matched to closing 1X2 market odds from `football-data.co.uk`. This case study is explicitly methodological, as with only 34 matches no statistically robust inference about long-run profitability can be drawn. The contribution is the demonstration of a complete pipeline from shot-level xG to match outcome probabilities, and a formal comparison of model calibration against market prices using Brier scores and log-likelihood comparisons.

The Poisson Framework.

Home and away xG are computed separately by summing shot-level predictions for each team in a match. Under the independent Poisson framework (Dixon and Coles, 1997), home goals H and away goals A are assumed to follow independent Poisson distributions:

$$H \sim \text{Poisson}(\lambda_h), \quad A \sim \text{Poisson}(\lambda_a) \quad (5.1)$$

where λ_h and λ_a represent the total home and away expected goals (xG), respectively. Match outcome probabilities are derived analytically across possible scorelines:

$$P(\text{Home Win}) = \sum_{i=0}^N \sum_{j=0}^{i-1} P(H = i) \cdot P(A = j) \quad (5.2)$$

$$P(\text{Draw}) = \sum_{i=0}^N P(H = i) \cdot P(A = i) \quad (5.3)$$

$$P(\text{Away Win}) = \sum_{i=0}^N \sum_{j=i+1}^N P(H = i) \cdot P(A = j) \quad (5.4)$$

with $N = 10$ as the upper truncation bound for goals. Market-implied probabilities are derived from closing average odds by removing the bookmaker overround via proportional normalization.

Match-Level xG Estimates.

The model predicts a mean home xG of 1.53 and away xG of 1.19 across the sample, closely matching StatsBomb’s benchmark estimates of 1.52 and 1.17, respectively. Both models underestimate actual realized goals scored (home: 1.77, away: 1.56). This discrepancy is consistent with the recognized tendency of shot-based xG aggregation to underrepresent match-level scoring variance, particularly in a small 34-match sample where high-scoring outliers meaningfully inflate the empirical mean above shot-by-shot expectations. The resulting Poisson-derived outcome probabilities are summarized in Table 5.10.

Table 5.10: Mean predicted probabilities vs. actual frequencies — Bundesliga 2023/24 ($n = 34$).

Outcome	Poisson Model	StatsBomb	Market Implied	Realized Frequency
Home Win	0.454	0.455	0.469	0.412
Draw	0.207	0.210	0.195	0.176
Away Win	0.339	0.335	0.336	0.412

Relative to realized match outcomes, all three probability sources overestimate home win frequency and draw frequency. The draw overestimation reflects a known property of independent Poisson models, which tend to inflate draw probabilities relative to empirical distributions when low-scoring dependence is unadjusted (Dixon and Coles, 1997). Mead et al. (2023) similarly report that match-outcome models rarely identify draws as the most probable result, as draws depend on the joint performance of both teams rather than on either team independently. Conversely, away wins occurred at an unusually high rate in this sample, an anomaly anticipated by neither xG aggregation nor market pricing.

Brier Score Comparison.

Table 5.11 evaluates forecast calibration using Brier Scores (BS), the mean squared error of probabilistic forecasts across all three sources. The Brier Skill Score (BSS) expresses each source’s calibration relative to market prices as a benchmark reference: $BSS = 1 - (BS_{\text{source}}/BS_{\text{market}})$, where positive values indicate superior calibration to the market.

Table 5.11: Brier scores by outcome and probability source (lower is better).

Source	Home Win BS	Draw BS	Away Win BS	Average BS	BSS vs Market
Our Model	0.1216	0.1466	0.1128	0.1270	-0.097
StatsBomb	0.1304	0.1527	0.1189	0.1340	-0.157
Market	0.1127	0.1389	0.0959	0.1158	—

Two primary findings emerge. First, the Poisson model built on open ElasticNet xG predictions achieves a lower average Brier score than StatsBomb, despite the latter’s superior AUC. Second, betting market prices outperform both xG-based models, yielding negative Brier Skill Scores. This market superiority aligns with established sports economics literature (Forrest and Simmons, 2000), as market prices incorporate broader contextual information beyond shot mechanics including team strength, roster availability, tactical match-ups, and market sentiment.

Likelihood Ratio & Model Fit Analysis.

To evaluate relative log-likelihood fit to observed match outcomes, log-likelihood values (ℓ) were computed for each probability source. The market price baseline achieved $\ell_{\text{market}} = -21.87$. The Poisson model yielded $\ell_{\text{model}} = -22.74$ ($\Delta\ell = -0.88$), while StatsBomb yielded $\ell_{\text{SB}} = -23.71$ ($\Delta\ell = -1.84$). Formal likelihood comparison confirms that shot-based xG probabilities do not provide a statistically significant improvement over closing market prices ($p > .05$), reflecting the efficient integration of match information in closing betting odds.

Correlation Analysis.

Model and market probabilities exhibit strong linear correlation for home wins ($r = 0.903$) and away wins ($r = 0.851$), suggesting that shot-based xG aggregation captures much of the outcome variation recognised by betting markets. The lower correlation for draws ($r = 0.402$) underlines the inherent difficulty of forecasting tie outcomes solely from expected goal totals, where small stochastic events heavily dictate the final scoreline.

Value Bet Simulation.

To illustrate practical betting applications, a simple value strategy was simulated by placing hypothetical unit bets whenever the model probability exceeded the market-implied probability.

Table 5.12: Value bet simulation results (Bundesliga 2023/24, $n = 34$ matches).

Strategy	Bets	Wins	Win %	Avg Odds	ROI
Bet Home Win	15	7	46.7%	3.341	-41.5%
Bet Draw	17	4	23.5%	5.342	+14.8%
Bet Away Win	15	6	40.0%	5.613	-27.4%
Combined	47	17	36.2%	—	-18.0%

Results are mixed across outcome types. Draw selections produced a positive return on investment, whereas home and away bets experienced negative returns, leading to a net ROI of -18.0% across 47 bets. Given the 34-match sample size, these simulation metrics lack statistical power to distinguish genuine edge from random variance and serve strictly as a methodological demonstration of pipeline execution.

Implications.

This case study demonstrates a complete, end-to-end pipeline connecting open 360 freeze-frame shot xG to match outcome probabilities. While shot-based xG aggregation captures much of the outcome variation recognised by betting markets, it does not match their absolute calibration accuracy, consistent with Baboota and Kaur (2019). Future extensions combining open

xG features with Dixon-Coles dependence parameters, team-level strength ratings, and multi-season datasets offer a clear path toward testing market efficiency in broader sports analytics applications.

Chapter 6

Conclusion

6.1 Summary of Findings and Concluding Remarks

This thesis examined whether spatiotemporal tracking data meaningfully improves expected goals prediction beyond standard event-data features, and whether this improvement depends on the choice of machine learning model. Both questions are answered affirmatively: spatiotemporal features provide genuine predictive gains, and the choice of model family materially determines how much of that gain is realised.

Tracking data provided the larger marginal lift over event-data-only specifications, while temporal possession-chain features contributed complementary but generally smaller gains. The statistical robustness of these gains was confirmed by paired DeLong tests with Holm-Bonferroni correction, demonstrating that all five model families showed significant AUC improvements from baseline to tuned specification, and that the tuned logistic regression was statistically indistinguishable from StatsBomb’s commercial benchmark. SHAP analysis revealed that temporal features dominated feature importance in the best-performing logistic regression, while tracking features dominated in Random Forest despite both using an identical feature set, indicating that the marginal value of each data stream depends materially on model architecture. A tuned ElasticNet logistic regression with the combined feature set was the best-performing model overall, achieving 98.0% of StatsBomb’s benchmark AUC. This study also introduces a novel set of temporal possession-sequence features, representing among the first bootstrap-validated possession-sequence predictors in the xG literature.

These findings suggest that transparency and interpretability need not come at a large cost to predictive accuracy, an appealing property given the general opacity of commercial xG providers.

For practitioners, freeze-frame tracking data delivered a larger and more consistent lift than temporal features, suggesting instantaneous positional data at the moment of the shot remains the more valuable acquisition relative to richer sequence-level context. Beyond prediction, the coefficient structure of the ElasticNet model has practical applications. The bootstrap-validated temporal coefficients, particularly ball carries, through balls, and cumulative chain pressure, suggest specific tactical contexts that generate higher quality shots, with implications for both player recruitment and tactical design.

More broadly, the result that a well-regularised linear model outperforms more complex alternatives given carefully engineered features echoes findings in the broader machine learning literature (Hastie et al., 2009), suggesting that investment in theoretically motivated feature engineering may yield greater returns than escalating model complexity when data is limited.

6.2 Limitations and Directions for Future Research

Data limitations. The absence of continuous tracking and vertical (z -axis) positional data in StatsBomb 360 freeze-frames precludes velocity and acceleration features, and is identified as the primary constraint on aerial modelling. The final dataset is modest relative to those in the literature, limiting statistical power in subgroup analyses. Future work with richer optical tracking data and continuous player kinematics would address the header modelling gap, while a larger and more competition-diverse dataset would enable formal subgroup inference and could explore continuous spatial representations and monotonic feature constraints, narrowing the gap to commercial benchmarks using open data.

Methodological limitations. The logistic regression interaction terms were selected based on theoretical motivation rather than systematic search, meaning potentially informative interactions may have been overlooked. Future specifications could employ cross-validated Lasso selection over a broader polynomial feature expansion to identify interactions in a more data-driven manner (Chen and Guestrin, 2016). AUC-ROC is used as the primary metric for model selection and comparison, consistent with the xG literature, though calibration and log loss are more directly relevant for practical probability estimation and are reported alongside AUC throughout Chapter 5. Isotonic regression calibration or Platt scaling could address the mild probability instability at high predicted xG values documented in Section 5.4.2.

Scope limitations. Player identity, goal differential, and other match-context features are deliberately excluded to preserve a clean shot-quality focus. This means the models likely

understate the ceiling of achievable predictive performance relative to studies incorporating such features. Consistent with the majority of the xG literature, only completed shot attempts are modelled, and the dataset’s five-competition coverage limits generalisation to a broader range of domestic leagues. Finally, the temporal possession framework introduced here could be extended to richer sequence representations using recurrent neural networks or transformer architectures, exploiting the full temporal structure of possession chains rather than summary statistics alone.

Bibliography

- Anthropic (2025). Claude (Version Sonnet 4.6). Large language model.
- Anzer, G. and Bauer, P. (2021). A goal scoring probability model for shots based on synchronized positional and event data in football (soccer). *Frontiers in Sports and Active Living*, 3:624475.
- Appiah, K. N., Adu, N., Singh, D. K., and Nartey, E. E. (2026). Machine learning expected goals models in soccer: Reproducible shot-outcome prediction with statsbomb data. Research Square preprint, Version 1. Posted 21 April 2026. Accessed August 2026.
- Baboota, R. and Kaur, H. (2019). Predictive analysis and modelling football results using machine learning approach for English Premier League. *International Journal of Forecasting*, 35(2):741–755.
- Bergstra, J. and Bengio, Y. (2012). Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, 13:281–305.
- Brechot, M. and Flepp, R. (2020). Dealing with randomness in match outcomes: How to rethink performance evaluation in european club football using expected goals. *Journal of Sports Economics*, 21(4):335–362.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.
- Caley, M. (2013). Shot matrix I: Shot location and expected goals. Cartilage Free Captain (online blog). Accessed August 2026.
- Cavus, M. and Biecek, P. (2022). Explainable expected goal models for performance analysis in football analytics. In *2022 IEEE 9th International Conference on Data Science and Advanced Analytics (DSAA)*, pages 1–9.
- Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of*

- the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pages 785–794.
- Corsaro, S., Dello Ioio, G., and Marino, Z. (2025). The evaluation of football players: An in-depth look at the expected goal metric. *Annals of Operations Research*. Online first.
- Davis, J. and Robberechts, P. (2024). Biases in expected goals models confound finishing ability.
- DeLong, E. R., DeLong, D. M., and Clarke-Pearson, D. L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics*, 44(3):837–845.
- Dixon, M. J. and Coles, S. G. (1997). Modelling association football scores and inefficiencies in the football betting market. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 46(2):265–280.
- Efron, B. and Tibshirani, R. J. (1993). *An Introduction to the Bootstrap*. Chapman and Hall, New York.
- Eggels, H., van Elk, R., and Pechenizkiy, M. (2016). Explaining soccer match outcomes with goal scoring opportunities predictive analytics. In *Proceedings of the 3rd Workshop on Machine Learning and Data Mining for Sports Analytics (MLSA 2016)*, Riva del Garda, Italy.
- Eggels, H. P. H. (2016). Expected goals in soccer: Explaining match results using predictive analytics. Master’s thesis, Eindhoven University of Technology.
- Fawcett, T. (2006). An introduction to ROC analysis. *Pattern Recognition Letters*, 27(8):861–874.
- FIFA (2024). FIFA annual report 2024. <https://inside.fifa.com/official-documents/annual-report/2024>. Accessed August 2026.
- Forrest, D. and Simmons, R. (2000). Forecasting sport: The behaviour and performance of football tipsters. *International Journal of Forecasting*, 16(3):317–331.
- Grand View Research (2025). Sports betting market size, share & trends analysis report. <https://www.grandviewresearch.com/industry-analysis/sports-betting-market-report>. Accessed August 2026.
- Grinsztajn, L., Oyallon, E., and Varoquaux, G. (2022). Why do tree-based models still outper-

- form deep learning on tabular data? *Advances in Neural Information Processing Systems*, 35.
- Hastie, T., Tibshirani, R., and Friedman, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, 2nd edition.
- Hewitt, J. H. and Karakuş, O. (2023). A machine learning approach for player and position adjusted expected goals in football (soccer). *Franklin Open*, 4:100034.
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.
- Lucey, P., Bialkowski, A., Monfort, M., Carr, P., and Matthews, I. (2015). “Quality vs quantity”: Improved shot prediction in soccer using strategic features from spatiotemporal data. In *Proceedings of the 9th MIT Sloan Sports Analytics Conference*.
- Lundberg, S. M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.
- Mead, J., O’Hare, A., and McMenemy, P. (2023). Expected goals in football: Improving model performance and demonstrating value. *PLOS ONE*, 18(4):e0282295.
- Méndez, M., Montero, C., and Núñez, M. (2023). Improving the expected goal value in football using multilayer perceptron networks. In *Recent Challenges in Intelligent Information and Database Systems (ACIIDS 2023)*, volume 1863 of *Communications in Computer and Information Science*, pages 352–363. Springer, Cham.
- Noordman, R. (2019). Improving the estimation of outcome probabilities of football matches using in-game information. MSc thesis, Econometrics, University of Amsterdam, Amsterdam School of Economics.
- Pollard, R., Ensum, J., and Taylor, S. (2004). Estimating the probability of a shot resulting in a goal: The effects of distance, angle and space. *International Journal of Soccer and Science*, 2(1):50–55.
- Pollard, R. and Reep, C. (1997). Measuring the effectiveness of playing strategies at soccer. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 46(4):541–550.
- Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., and Gulin, A. (2018). CatBoost:

- Unbiased boosting with categorical features. *Advances in Neural Information Processing Systems*, 31.
- Rathke, A. (2017). An examination of expected goals and shot efficiency in soccer. *Journal of Human Sport and Exercise*, 12(2proc):S514–S529.
- Scholtes, A. and Karakuş, O. (2024). Bayes-xG: Player and position correction on expected goals (xG) using bayesian hierarchical approach. *Frontiers in Sports and Active Living*, 6:1348983.
- Shwartz-Ziv, R. and Armon, A. (2022). Tabular data: Deep learning is not all you need. *Information Fusion*, 81:84–90.
- Spearman, W. (2018). Beyond expected goals. In *Proceedings of the 12th MIT Sloan Sports Analytics Conference*, pages 1–17.
- StatsBomb (2022). Expected goals. Technical report, StatsBomb. Summer 2022 methodology documentation.
- StatsBomb (2024). StatsBomb expected goals model. <https://statsbomb.com/>. Accessed August 2026.
- Wolpert, D. H. (1992). Stacked generalization. *Neural Networks*, 5(2):241–259.

Appendix A: Feature Descriptions and Temporal Statistics

Table 1: Variable descriptions for key features referenced in Figures 3.1 and 3.2, and Table 5.6.

Variable	Description
distance	Euclidean distance from shot to goal centre (m)
angle_rad	Angle subtended by the goalposts from the shooter (rad)
y_centrality	Deviation from the goal's central axis (m)
body_part_prob	Target-encoded conversion rate for shot body part
shot_technique_Normal	Indicator for standard (non-volley) shot technique
gk_off_line	Goalkeeper's distance from the goal line (m)
gk_dist	Distance between goalkeeper and shooter (m)
gk_angle_error	Goalkeeper's angular deviation from the shot line (rad)
nearest_opp_dist	Distance to the nearest defender (m)
cone_opponents	Defenders within the shooter-to-goalposts triangle
defenders_in_box	Opponents within the penalty area
num_opponents	Total opponents visible in the freeze frame
no_blocker_x_angle	Unobstructed shot path $\times$ shot angle
chain_has_throughball	Possession chain included a through ball
chain_under_pressure	Pressure events faced during the possession chain
w_carries	Ball carries in the 10s window before the shot
w_under_pressure	Pressure events in the 10s window before the shot
carry_into_box	A carry entered the penalty box during the chain
defensive_line_x	Mean x-coordinate of the defensive line at the shot
def_line_x_distance	Defensive line x-coordinate $\times$ shot distance
prog_dist_x_gk_off_line	Chain progressive distance $\times$ GK off-line distance

Temporal Feature Descriptive Statistics

Table 2: Full temporal feature comparison, goals vs non-goals.

Feature	Non-Goal	Goal	Diff%
w_passes	2.240	2.495	+11.4%
w_carries	1.970	2.162	+9.7%
w_pressure_events	1.008	1.077	+6.8%
w_speed	1.918	2.287	+19.2%
w_prog_distance	17.227	20.695	+20.1%
chain_passes	7.227	7.040	−2.6%
chain_duration	24.733	23.367	−5.5%
chain_pressures	2.284	2.026	−11.3%
chain_has_cross	0.281	0.313	+11.4%
chain_has_throughball	0.036	0.115	+219.4%
chain_prog_distance	39.497	48.017	+21.6%
fast_break	0.017	0.021	+23.5%
pass_into_box	0.598	0.659	+10.2%
carry_into_box	0.346	0.425	+22.8%
n_defenders_beaten	0.164	0.181	+10.4%
final_pass_length	22.499	23.518	+4.5%
shot_from_rebound	0.191	0.196	+2.6%
defensive_line_x	104.460	106.508	+2.0%

Table 3: Temporal feature descriptive statistics.

Statistic	w_passes	w_carries	w_speed	w_prog_distance	chain_passes
Mean	2.264	1.988	1.954	17.560	7.209
Median	2.000	2.000	1.182	10.300	5.000
Std	1.231	1.391	2.542	22.119	7.833
Min	0.000	0.000	0.000	0.000	0.000
Max	8.000	8.000	23.446	116.300	76.000

Statistic	chain_duration	chain_prog_distance	final_pass_length	defensive_line_x
Mean	24.601	40.315	22.597	104.657
Median	16.421	40.800	19.587	105.213
Std	25.411	37.832	15.896	6.073
Min	0.000	-89.200	0.000	30.545
Max	222.437	114.400	102.020	119.331

Table 4: Binary temporal features and goal conversion rates. Overall conversion rate: 9.6%.

Feature	n (=1)	Conv. (=0)	Conv. (=1)	Ratio
chain_has_throughball	343	0.089	0.254	2.86×
chain_has_cross	2,243	0.092	0.106	1.15×
chain_has_switchplay	1,453	0.098	0.086	0.88×
fast_break	137	0.096	0.117	1.22×
pass_into_box	4,758	0.083	0.105	1.27×
carry_into_box	2,788	0.085	0.115	1.35×
shot_from_rebound	1,507	0.095	0.098	1.03×

Table 5: Temporal features by play pattern.

Play Pattern	Shots	Conv.%	Avg. Speed	Avg. Carries	Avg. Duration	Avg. Passes	%Through
From Counter	354	13.3	3.068	2.107	15.546	3.511	8.5
From Keeper	132	12.9	2.158	2.371	36.405	9.826	7.6
From Goal Kick	343	11.7	2.629	2.411	32.758	9.563	4.7
From Throw In	1,545	10.1	2.219	2.447	28.036	8.539	4.7
Regular Play	2,647	9.8	2.674	2.597	30.598	8.711	5.2
From Corner	1,439	8.5	0.577	0.732	12.187	3.818	1.7
From Free Kick	1,333	8.1	1.186	1.408	20.882	6.292	3.5
From Kick Off	92	7.6	2.321	2.261	29.949	9.685	7.6

Appendix B: Supplementary Results

Table 6: Full model leaderboard across all feature conditions. Tuned models use the full feature set. All other models are untuned.

Model	AUC	Log Loss
StatsBomb	0.867	0.228
<i>Logistic Regression</i>		
LR Tuned	0.850	0.243
LR Tracking	0.841	0.243
LR T+T	0.832	0.246
LR Temporal	0.832	0.246
LR Baseline	0.826	0.254
<i>Random Forest</i>		
RF Tuned	0.846	0.245
RF T+T	0.842	0.246
RF Tracking	0.840	0.245
RF Temporal	0.834	0.253
RF Baseline	0.791	0.262
<i>LightGBM</i>		
LGB Tuned	0.830	0.248
LGB Tracking	0.804	0.266
LGB T+T	0.801	0.277
LGB Temporal	0.794	0.275
LGB Baseline	0.740	0.294
<i>XGBoost</i>		
XGB Tuned	0.835	0.246
XGB Tracking	0.817	0.251
XGB T+T	0.816	0.255
XGB Temporal	0.810	0.255
XGB Baseline	0.782	0.264
<i>CatBoost</i>		
CB T+T	0.838	0.246
CB Tuned	0.830	0.248
CB Tracking	0.821	0.249
CB Temporal	0.817	0.256
CB Baseline	0.796	0.262

Table 7: Paired DeLong tests: tuned models vs StatsBomb benchmark. Holm-Bonferroni correction applied across five simultaneous comparisons.

Model	Model AUC	StatsBomb AUC	Δ AUC	Holm p	Significant
LR Tuned	0.850	0.867	-0.017	0.095	No
RF Tuned	0.846	0.867	-0.021	0.095	No
XGB Tuned	0.835	0.867	-0.032	0.009	Yes
CB Tuned	0.830	0.867	-0.037	0.008	Yes
LGB Tuned	0.830	0.867	-0.037	0.004	Yes

Paired DeLong test (DeLong et al., 1988). Δ AUC = Model AUC - StatsBomb AUC. Holm-Bonferroni corrected p-values reported.

Table 8: Neural network (MLP) results across feature conditions. Tuned via randomised search (15 iterations, 3-fold CV).

Feature Condition	AUC
StatsBomb benchmark	0.867
NN — Full (tuned)	0.832
NN — Tracking (tuned)	0.835
NN — Baseline (tuned)	0.803